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NMR SPECTROSCOPY IN THE STRUCTURAL ELUCIDATION OF OLIGOSACCHARIDES AND GLYCOSIDES*†

PAWAN K. AGRAWAL

Central Institute of Medicinal and Aromatic Plants, Lucknow 226 016, India

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Abstract—The potential of one- and two-dimensional NMR techniques for the identification of individual sugar residues, their anomeric configuration, interglycosidic linkages, sequencing and the site of any appended group, in establishing the structures of naturally occurring oligosaccharides and glycosides is presented.

INTRODUCTION

Carbohydrates are an integral constituent of all living organisms and are associated with a variety of vital functions which sustain life. These are called homopolysaccharides, or homopolysaccharides, when only one type of monosaccharide unit is present, such as in starch and cellulose, and heteropolysaccharides or heteropolysaccharides when more than one kind of monosaccharide is the constituent unit, such as in gums, mucillages, pectins and hemicelluloses, etc. In addition to their occurrence in polymeric forms, [1–4] carbohydrates occur quite frequently either in free monomeric and oligomeric forms or in *O*- and/or *C*-glycosidic form of some non-sugar moiety; most naturally occurring compounds also exist in glycosidic form. These are found most abundantly in seaweeds, fungi and higher terrestrial plants [5–17]. Many oligosaccharides and glycosides exhibit potent biological activity [5–17]. The determination of an exact structure is often difficult, sometimes even for a monosaccharide, because many carbohydrates differ only in their stereochemistry, resulting in highly similar spectral data. Thus, structure establishment of an oligosaccharide is not an easy task because of the close similarity in the structures of constituent sugar residues and due to the existence of multiple substitution points.

To solve the complete structure of an oligosaccharide, the following questions must be addressed: (a) What is the monosaccharide composition? (b) What are the anomeric configurations of each glycosidically-linked monosaccharide unit? (c) How are the monosaccharide units linked to one another? and (d) What are the appended groups, if any? However, if the structure of a glycoside is

under consideration, then in addition to all of the above-mentioned questions to be answered, one needs to characterize the aglycone residue followed by the establishment of the site of attachment of the sugar residues to it. No doubt, the structure of the intact glycoside can be established by NMR spectroscopic methods but the structure of the aglycone may be quite variable. Therefore, emphasis has been provided herein to the structure establishment of the sugar portion only. However, it is considered worthwhile to present a brief discussion of the determination of the site of glycosidic linkage in glycosides. Rather than providing a comprehensive coverage of examples scattered in the literature, the usefulness of various NMR techniques in solving the structures of oligosaccharides and glycosides is presented.

Several experimental methods have been applied to determine the chemical structure of an oligosaccharide or glycoside, but the most common procedure is the analysis of fragments obtained by chemical and enzymatic degradation. Among the classical methods, methylation analysis [18, 19] involves tedious chemistry and its sensitivity is unimpressive. Derivatization via permethylation, peracetylation or trimethylsilylation followed by mass spectrometry has been widely used for structural analysis of natural carbohydrates [20]. Furthermore, methylation analysis by GC-MS [21] yields information only on positions of substitution and not on the anomeric configuration or sequence of monosaccharides in an oligosaccharide.

Permethylation followed by hydrolysis experiments to determine the sugar sequence and the site of interglycosidic linkage has been widely employed for the structure establishment of saponins having triterpenoids, steroids or steroidal alkaloids as the sapogenin moiety [22]. Hydrolysis and other cleavage methods amenable to oligosaccharide analysis have been recently reviewed [23]. This whole process again is somewhat tedious and also consumes oligosaccharide, which is often very difficult to separate and purify and may be better employed in biological evaluation experiments.

*Dedicated to Dr R. P. Rastogi, on the occasion of his 69th birthday.

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Recently, a reductive cleavage method has been introduced to assist in primary structure elucidation [24, 25]. Although enzymatic digestion appears to be a valuable method, the number of known exoglycosidases is quite small. The availability of endoglycosidases whose specificity is well documented remains very limited. Mass spectrometry, especially field desorption (FD) [26], fast atom bombardment (FAB) [27, 28], plasma desorption (PD) and high performance tandem mass spectrometry seem to be very attractive on account of their speed and sensitivity [29, 30]. All of these methods may give limited partial structural information. When the sample quantities are limited, however, one must rely on such methods.

Of all the modern structural methods for oligosaccharides, NMR spectroscopy yields the most complete picture of oligosaccharide structure and behaviour in solution with or without prior structural knowledge (Table 1). It is the only method which can, in principle, give an *ab initio* structure without resort to any other method [31–33]. In practice, a complete structural determination by NMR is really the best approach to a completely new carbohydrate structure and generally other methods are used in conjunction with NMR. In addition to determining primary structure, NMR provides information on the conformation and molecular dynamics (motional characteristics) of the molecule in a solution state.

The major weakness of NMR spectroscopy as a method for structure determination is its poor sensitivity. On the other hand, since the experiment is non-destructive, it should always be considered first after the isolation of a suitable sample. A NMR spectrum (^1H and/or ^{13}C) is a modest experimental effort and will give immediate information on the purity of the sample and perhaps some general information on its structure (Table 2). In the most favourable cases, the structure can be completely determined by this simple experiment. In the worst case only time is lost and because of the non-invasive nature of NMR methods, the intact material can generally be easily recovered by removal of solvent for subsequent analysis

by methylation, enzymatic degradation or mass spectrometry, all of which are destructive.

SOLVENT AND TEMPERATURE

Although most oligosaccharides and glycosides are soluble in D_2O , many workers have shown that glycosides give well resolved NMR spectra in non-aqueous solvents, such as $\text{DMSO-}d_6$ and $\text{pyridine-}d_5$. Chemical shifts are reported relative to internal TMS and they differ somewhat from those of the same carbohydrate in aqueous solution. The structural reporter group (a group which give signals outside the bulk region in ^1H NMR spectra) approach [34] is quite useful for correlation of carbohydrate structure because the chemical shifts of these groups provide valuable structural information. Chemical shift analogies between spectra of oligosaccharide in $\text{DMSO-}d_6$ or $\text{pyridine-}d_5$ with the spectra of oligosaccharides in D_2O cannot be effectively drawn due to not only the difference in the chemical shift reference but also to other perturbations in the chemical shifts due to the varying magnitude of solute–solvent interactions. The ^1H and ^{13}C NMR chemical shifts of some frequently occurring monosaccharides are presented in Tables 3 and 4. Generally, ^1H NMR spectra obtained in D_2O are referenced to internal acetone (2.225 ppm at 25°).

It should also be mentioned here that differential isotope shifts ($\delta_{\text{O-deuterated}} - \delta_{\text{O-undeuterated}}$) measured in $\text{pyridine-}d_5$ are useful not only for the assignment of ^{13}C resonances but also for the establishment of glycosidic linkages. The ^{13}C resonance with free hydroxyl groups shifts to lower field under these solvent conditions whereas those carbon resonances which lack a free OH group, for example, those involved in glycosidic linkage, exhibit negligible effects [35, 36].

In earlier studies, hydroxyl groups of sugar residues were most often converted to their methyl or trimethylsilyl ethers to eliminate peaks due to hydroxyl groups from the spectrum. But now this is achieved by D_2O

Table 1. Structural information and NMR methods

Structural information	NMR methods
1. Number of sugar residues	a. Integrated 1D ^1H NMR spectrum b. ^{13}C NMR spectrum c. 2D ^1H – ^1H correlation spectroscopy for connectivity analysis d. 2D ^1H – ^{13}C correlation spectroscopy
2. Constituent monosaccharides	a. ^1H NMR chemical shifts b. ^1H NMR vicinal coupling constants ($^3J_{\text{H,H}}$) c. 2D homonuclear correlation spectroscopy (COSY, HOHAHA) d. ^{13}C NMR chemical shifts e. ^1H – ^{13}C correlation spectroscopy
3. Anomeric configuration	a. ^1H NMR chemical shifts and vicinal coupling constants b. ^{13}C NMR chemical shifts and ^{13}C – ^1H coupling constants c. Intraresidue NOE
4. Linkage sites and sequence	a. ^1H and ^{13}C NMR chemical shifts b. Interresidue NOE c. Long-range homo- and heteronuclear correlation
5. Position of appended groups	a. ^1H and ^{13}C NMR chemical shifts b. Interresidue NOE c. Long-range homo- and heteronuclear correlation

Table 2. Representative ^1H and ^{13}C NMR chemical shifts for oligosaccharides*

^1H	ppm	^{13}C	ppm
MeC	1.1–1.3	MeC	16–18
MeCON	2.0–2.2	MeCON	22–23.5
MeCOO	1.8–2.2	MeCOO	18–22
CH(NH)	3.0–3.8	CH(NH)	52–58
MeO	3.3–3.5	MeO	55–61
H-2 to H-6†	3.2–4.5	CH ₂ OH	57.7–64.7
		CH ₂ OR‡	66–70
		C-2 to C-5	65–87
H-1 (ax)	4.3–4.8	C-1 (ax-O, red)	90–95
		C-1 (ketoses)	98–100
		C-1 (ax-O, glyc)	98–103
H-1 (eq)	5.1–5.8	C-1 (eq-O, red)	95–98
		C-1 (eq-O, glyc)	103–106
		C-1 (fur)	103–112
		COO	170–180

*Unless otherwise stated ^1H and ^{13}C data represent pyranose form. Abbreviations: red, reducing end sugar; glyc, glycosidically linked sugar; fur, furanose; ax, axial; eq, equatorial.

†In unacylated as well as C-6 acylated.

‡In C-6 glycosylated.

exchange of the sample by repeated evaporation or lyophilization from D_2O prior to dissolving the sample in any good-quality deuterated solvent if determination of the chemical shifts of the exchangeable protons (OH and/or NH) is not the aim, as these on the one hand provide some valuable information, and on the other hand complicate the ^1H NMR spectrum. This exchange substitutes the exchangeable protons in the sample with deuterium and thus eliminates their signals in ^1H NMR spectra. Such exchange is not essential when measurement of the ^{13}C NMR spectrum is the primary objective.

Although the chemical shifts of a few of the protons are slightly temperature-dependent, the effect is usually small. When a small quantity of oligosaccharide is available for study, the HOD signal can pose a major problem. In such cases, the temperature can be adjusted to move the resonance of the residual HOD line so that it does not obscure any sugar resonances. By raising the temperature, one can displace the HOD signal upfield (ca 4.5 ppm at 70°) thereby exposing more of the spectrum. The solvent signal can also be eliminated by a number of FT techniques [37]. Sometimes, ^1H NMR spectra of oligosaccharides show considerable line broadening at ambient temperature and, under these circumstances, well resolved lines can be observed due to segmental motion, especially if spectra are measured at elevated temperatures.

MONOSACCHARIDE COMPOSITION AND ANOMERIC CONFIGURATION

The chemical shifts and coupling constant information available from ^1H and ^{13}C NMR spectra are of significance in obtaining this information. In some cases comparison of ^1H and ^{13}C NMR spectra of an unknown oligosaccharide with spectral data found in the literature for related oligosaccharides or glycosides [38–43] can

provide relevant information about its monosaccharide composition.

ONE-DIMENSIONAL NMR METHODS

The measurement of a one-dimensional ^1H and a ^{13}C NMR spectrum is the first step in getting primary structural information. This is because ^1H and ^{13}C chemical shifts of the resonances of a monosaccharide residue within a polysaccharide depends mainly upon the structure of monosaccharides and upon the nature of the flanking sugar residues.

Reasonable ^1H NMR spectra can be obtained with ca 1 mg of an oligosaccharide but relatively large sample amounts, typically 5–10 mg, or longer acquisition times for still smaller amounts are required for measurement of ^{13}C NMR spectra. Whatever the case it is desirable to measure both spectra but at least a ^1H NMR spectrum if it is not possible to get a ^{13}C NMR spectrum. In such cases, inverse-detected techniques (*vide infra*) are useful as these provide heteronuclear correlation maps. However, it should be mentioned that both are important tools for structural characterization of carbohydrates and their derivatives.

^1H NMR methods. The 1D ^1H NMR spectrum of an oligosaccharide shows only some recognizable signals, such as anomeric protons at 4.3–5.9 ppm, methyl doublets of 6-deoxy sugar residues at 1.1–1.3 ppm, methyl singlets of acetamido groups at 2.0–2.2 ppm and various other protons with distinctive chemical shifts. These, in conjunction with vicinal ^1H – ^1H coupling constants ($^3J_{\text{HH}}$) can be correlated with known structures to yield relevant information in terms of primary structure [44]. Signal line-widths and spectral integrations also identify the types of monosaccharide units present and their relative abundances. The relative intensities of isolated resonances can be used as markers for the establishment of the purity of the compound. Often from the spectrum, it can be deduced whether or not the sample consists of more than one carbohydrate structure, and if so, what compositional ratios of components are present in the mixture. The vast majority of proton resonances appear in a very small spectral width of 3.0–4.2 ppm, with subsequent overlap problems. These derive from the bulk of nonanomeric sugar methine and methylene protons which have very similar chemical shifts in different monosaccharide residues. Significant overlap makes it difficult to locate the resonances of individual nuclei and to assign these resonances to a specific monosaccharide residue. The two main problems are the presence of more than one form of the reducing end sugar at equilibrium, and the relatively small range of chemical shifts for the non-anomeric protons, leading to a crowded and a strongly coupled spectrum. In addition, even when all the parameters of the spin system can be extracted from the spectrum, there remain assignment problems, since many proton–proton coupling constants (vicinal and geminal) are very similar. Therefore, the first task is to perform a through-bond connectivity analysis in order to determine the number of different spin systems corresponding to individual sugar residues that are either different themselves or are the same but located in a different environment. Following this, the presence of multiple identical sugar units situated in different environments can be demonstrated by integrating well-resolved signals, usually those of anomeric protons in the 1D spectrum. However, in

Table 3. ¹H NMR data for glycopyranoses and methyl glycopyranosides

Sugar	H-1	H-2	H-3	H-4	H-5	H-6	MeCONH
Glycopyranoses							
β-D-Glc	4.64	3.25	3.50	3.42	3.46	3.72, 3.90	—
α-D-Glc	5.23	3.54	3.72	3.42	3.84	3.76, 84	—
β-D-Gal	4.53	3.45	3.59	3.89	3.65	3.64, 3.72	—
α-D-Gal	5.22	3.78	3.81	3.95	4.03	3.69, 3.69	—
β-D-Man	4.89	3.95	3.66	3.60	3.38	3.75, 3.91	—
α-D-Man	5.18	3.94	3.86	3.68	3.82	3.74, 3.86	—
β-L-Rha	4.85	3.93	3.59	3.38	3.39	1.30	—
α-L-Rha	5.12	3.92	3.81	3.45	3.86	1.28	—
β-L-Fuc	4.55	3.46	3.63	3.74	3.79	1.26	—
α-L-Fuc	5.20	3.77	3.86	3.81	4.20	1.21	—
β-D-GlcNAc	4.72	3.65	3.56	3.46	3.46	3.75, 3.91	2.06
α-D-GlcNAc	5.21	3.88	3.75	3.49	3.86	3.77, 3.85	2.06
β-D-GalNAc	4.68	3.90	3.77	3.98	3.72	3.82, 3.84	2.06
α-D-GalNAc	5.28	4.19	3.95	4.05	4.13	3.79, 3.79	2.06
β-D-ManNAc	5.01	4.45	3.83	3.52	3.45	3.81, 3.90	2.06
α-D-ManNAc	5.13	4.31	4.07	3.63	3.86	3.84, 3.84	2.10
β-D-GlcA-Na	4.65	3.30	3.52	3.54	3.72	—	—
α-D-GlcA-Na	5.24	3.59	3.75	3.53	4.09	—	—
β-D-GalA-Na	4.56	3.51	3.69	4.23	4.03	—	—
α-D-GalA-Na	5.30	3.83	3.92	4.29	4.39	—	—
β-D-ManA-Na	4.89	3.93	3.66	3.72	3.63	—	—
α-D-ManA-Na	5.22	3.90	3.87	3.83	4.04	—	—
Methyl glycopyranosides							
β-D-Glc	4.27	3.15	3.38	3.27	3.36	3.82, 3.62	—
α-D-Glc	4.70	3.46	3.56	3.29	3.54	3.77, 3.66	—
β-D-Gal	4.20	3.39	3.53	3.81	3.57	3.69, 3.64	—
α-D-Gal	4.73	3.72	3.68	3.86	3.78	3.67, 3.61	—
β-D-Qui	4.25	3.15	3.33	3.04	3.38	1.19	—
α-D-Qui	4.64	3.47	3.51	3.04	3.61	1.17	—
β-D-6-Dgal	4.19	3.36	3.52	3.62	3.69	1.15	—
α-D-6-Dgal	4.64	3.67	3.70	3.68	3.92	1.11	—
β-D-6-Dman	4.43	3.87	3.48	3.26	3.29	1.21	—
α-D-6-Dman	4.59	3.82	3.60	3.33	3.56	1.19	—
β-D-Man	4.47	3.88	3.53	3.46	3.27	3.83, 3.63	—
α-D-Man	4.66	3.82	3.65	3.53	3.51	3.79, 3.65	—
β-D-Ara	4.72	3.74	3.72	3.89	3.55, 3.77	—	—
α-D-Ara	4.16	3.43	3.57	3.85	3.82, 3.57	—	—
β-D-Rib	4.52	3.51	3.91	3.79	3.74, 3.61	—	—
α-D-Rib	4.51	3.70	3.86	3.72	3.47, 3.68	—	—
β-D-Xyl	4.21	3.14	3.33	3.51	3.88, 3.21	—	—
α-D-Xyl	4.67	3.44	3.53	3.47	3.59, 3.39	—	—

Table 4. ¹³C NMR data for glycopyranose and methyl glycopyranosides

Sugar	C-1	C-2	C-3	C-4	C-5	C-6	MeCONH
Glycopyranoses							
β-D-Glc	96.8	75.2	76.7	70.7	76.7	61.8	—
α-D-Glc	93.0	72.4	73.7	70.7	72.3	61.8	—
β-D-GlcN	93.6	57.9	72.9	70.6	76.9	61.4	—
α-D-GlcN	89.9	55.5	70.6	70.5	72.4	61.3	—
β-D-Gal	97.4	72.9	73.8	69.7	75.9	61.8	—
α-D-Gal	93.2	69.3	70.1	70.3	71.3	62.0	—
β-D-Man	94.5	72.1	74.0	67.7	77.0	62.0	—
α-D-Man	94.9	71.7	71.2	67.9	73.3	62.0	—
β-L-Rha	94.4	72.2	73.8	72.8	72.8	17.6	—
α-L-Rha	94.8	71.8	71.0	73.2	69.1	17.7	—
β-L-Fuc	97.2	72.7	73.9	72.4	71.6	16.3	—
α-L-Fuc	93.1	69.1	70.3	72.8	67.1	16.3	—

Table 4. *Continued*

Sugar	C-1	C-2	C-3	C-4	C-5	C-6	MeCONH
β -D-GlcNAc	95.9	57.9	74.8	71.1	76.8	61.9	23.1, 175.5
α -D-GlcNAc	91.8	55.0	71.7	71.3	72.5	61.8	22.9, 175.1
β -D-GalNAc	96.3	54.8	72.0	68.9	76.0	61.9	23.1, 175.8
α -D-GalNAc	92.0	51.2	68.4	69.6	71.4	62.1	22.9, 175.4
β -D-ManNAc	93.9	54.9	73.0	67.7	77.3	61.5	23.0, 176.4
α -D-ManNAc	94.0	54.1	69.8	67.9	73.0	61.5	22.8, 175.4
β -D-GlcA-Na	96.8	75.0	76.5	72.7	76.9	176.5	—
α -D-GlcA-Na	93.0	72.3	73.5	72.9	72.5	177.4	—
β -D-GalA-Na	96.9	72.6	73.8	71.2	76.4	175.6	—
α -D-GalA-Na	93.1	69.0	70.3	71.6	72.3	176.4	—
β -D-ManA-Na	94.5	71.9	73.9	69.6	77.0	176.8	—
α -D-ManA-Na	94.8	71.4	71.1	70.0	73.7	177.8	—
α -D-All	94.3	72.2	72.0	67.7	74.4	62.1	—
α -D-All	93.7	67.9	72.0	66.9	67.7	61.6	—
β -D-Ara	93.4	69.5	69.5	69.5	63.4	—	—
α -D-Ara	97.6	72.9	73.5	69.6	67.2	—	—
β -D-Rib	94.7	71.9	69.7	68.2	63.8	—	—
α -D-Rib	94.3	70.8	70.1	68.1	63.8	—	—
β -D-Xyl	97.5	75.1	76.8	70.2	66.1	—	—
α -D-Xyl	93.1	72.5	73.9	70.4	61.9	—	—
β -L-Tal	95.0	72.5	69.6	69.4	76.5	62.2	—
α -L-Tal	95.5	71.7	66.0	70.6	72.0	62.4	—
β -D-Sor	64.6	99.5	70.2	73.6	71.1	59.8	—
α -L-Sor	64.5	98.6	71.4	74.8	70.3	62.7	—
β -D-Fru	64.8	99.0	68.5	70.6	70.1	64.2	—
α -D-Fru	65.9	99.1	70.9	71.3	70.0	61.9	—
β -D-Tag	64.4	99.1	64.6	70.7 ^a	70.1 ^c	61.0	—
α -D-Tag	64.8	99.0	70.8 ^a	70.7 ^a	67.2	63.1	—
β -D-ManNAc	93.9	54.9	73.0	67.7	77.3	61.5	23.0, 176.4
α -D-ManNAc	94.0	54.1	69.8	67.9	73.0	61.5	22.8, 175.4
β -D-GlcA-Na	96.8	75.0	76.5	72.7	76.9	176.5	—
α -D-GlcA-Na	93.0	72.3	73.5	72.9	72.5	177.4	—
β -D-GalA-Na	96.9	72.6	73.8	71.2	76.4	175.6	—
α -D-GalA-Na	93.1	69.0	70.3	71.6	72.3	176.4	—
β -D-ManA-Na	94.5	71.9	73.9	69.6	77.0	176.8	—
α -D-ManA-Na	94.8	71.4	71.1	70.0	73.7	177.8	—
Methyl glycopyranosides							
β -D-Glc	104.0	74.1	76.8	70.6	76.8	61.8	—
α -D-Glc	100.0	72.2	74.1	70.6	72.5	61.6	—
β -D-GlcN	93.6	57.9	72.9	70.6	76.9	61.4	—
α -D-GlcN	89.9	55.5	70.6	70.5	72.4	61.3	—
β -D-Gal	104.5	71.7	73.8	69.7	76.0	62.0	—
α -D-Gal	100.1	69.2	70.5	70.2	71.6	62.2	—
β -D-Man	102.3	71.7	74.5	68.4	77.6	62.6	—
α -D-Man	102.2	71.4	72.1	68.3	73.9	62.5	—
β -L-Rha	102.4	71.8	74.1	73.4	73.4	17.9	—
α -L-Rha	102.1	71.2	71.5	73.3	69.5	17.9	—
β -L-Fuc	97.2	72.7	73.9	72.4	71.6	16.3	—
α -L-Fuc	93.1	69.1	70.3	72.8	67.1	16.3	—
β -D-GlcNAc	102.5	57.0	74.7	70.8	76.5	61.6	23.1, 175.3
α -D-GlcNAc	98.9	54.4	72.2	70.6	72.1	61.4	22.8, 175.1
β -D-GalNAc	96.3	54.8	72.0	68.9	76.0	61.9	23.1, 175.8
α -D-GalNAc	92.0	51.2	68.4	69.6	71.4	62.1	22.9, 175.4
β -D-ManNAc	93.9	54.9	73.0	67.7	77.3	61.5	23.0, 176.4
α -D-ManNAc	94.0	54.1	69.8	68.0	73.0	61.5	22.8, 175.4
β -D-GlcA-Na	96.8	75.0	76.5	72.7	76.9	176.5	—
α -D-GlcA-Na	93.0	72.3	73.5	72.9	72.5	177.4	—
β -D-GalA-Na	96.9	72.6	73.8	71.2	76.4	175.6	—
α -D-GalA-Na	93.1	69.0	70.3	71.6	72.3	176.4	—
β -D-ManA-Na	94.5	71.9	73.9	69.6	77.0	176.8	—
α -D-ManA-Na	94.8	71.4	71.1	40.0	73.7	177.8	—

^1H NMR spectra, the limited chemical shift range, and the presence of homonuclear ^1H – ^1H spin-coupling, often conspire to create non-first order spectra which cannot be easily interpreted to yield valid ^1H – ^1H spin couplings without the need for spectral simulation by computer.

The isolated ^1H NMR resonances, resonating at uncrowded regions of the spectrum, called 'structural reporter resonances' [34], generally act as the starting points. This means that the chemical shifts of protons resonating at clearly distinguishable positions in the spectrum, together with their coupling constants and line width, provide information essential to assignment of the structure. These include those of the methyl groups (H-6) of 6-deoxy sugars and the anomeric resonances (H-1). The chemical shifts of some of these resonances may be characteristic of the position of the glycosidic linkage. Most of the structural reporter resonances appear in the low field region between 4.2 and 5.3 ppm. Usually the anomeric resonances of α -glycosides resonate at a downfield position by 0.3–0.5 ppm compared with that of the corresponding β -glycosides. Thus, resonances at lowest field (4.8–5.3 ppm) which are doublets with $^3J_{1,2}$ in the range 1–4 Hz are those of α -anomeric protons, whereas β -anomeric protons appear as doublets between 4.4–4.8 ppm with $^3J_{1,2}$ in the range 6–8 Hz in monosaccharides with *gluco* and *galacto* stereochemistry.

The value of the structural reporter method is in its simplicity, since only a simple 1D ^1H NMR spectrum is needed. However, its weakness is the requirement for closely related carbohydrate structures in order to reach unambiguous conclusions. When a novel structure is under investigation model compounds are unlikely to be available, and therefore assignment of protons is consequently more difficult. Thus, it is the method of choice when dealing with members of a class which have been previously studied in detail by ^1H NMR.

In pyranosides, the six-membered ring generally forms a chair of fixed conformation providing a classification of protons as axial or equatorial. Therefore, the coupling patterns are characteristic of the stereochemistry of type of the carbohydrate. For example, if the H-2 is axial, as it is for *gluco* and *galacto* stereochemistry, then a small coupling constant ($^3J_{\text{HH}}$) of ca 2–4 Hz is observed as a result of the *gauche* conformation of H-1 and H-2 following the Karplus relation (dihedral angle ca 60°). The *trans* diaxial relationship of H-1 and H-2 in β -anomers of sugars with a *gluco* and *galacto* configuration leads to larger (7–9 Hz) coupling constants (dihedral angle ca 180°). Chemical shifts between 4.4 and 4.8 ppm are typical of the anomeric protons of these β -linked residues, whereas α -anomeric protons usually resonate between 4.9–5.3 ppm. The equatorial orientation of H-2 as in mannose results in a small dihedral angle and thus a small $^3J_{\text{HH}}$ for both α - and β -anomers, therefore making assignment of the anomeric configuration more difficult.

Information about signal, assignments for an oligosaccharide can be obtained by comparison with model compounds, particularly with monosaccharides (Table 3), or with oligosaccharides with fewer monosaccharide residues. These data form the groundwork for the sequence determination of an oligosaccharide. Various 1D methods such as spin decoupling, NOE (nuclear Overhauser effect), INDOR (Internuclear Double resonance) and partially relaxed spectroscopy have been used to unravel hidden resonances in the unresolved envelope [45]. Once individual resonances have been assigned to

specific sugar residues, then NOE and relaxation experiments involving these resonances can help to determine linkage and sequence [46–52], but substantial overlap of multiplets in the region 3.4–4.2 ppm generally prohibits unambiguous assignment of ^1H resonances to individual residues.

^{13}C NMR methods. While ^1H NMR spectroscopy has been the most important source of structural information, ^{13}C NMR spectroscopy also has enormous potential for carbohydrates and glycosides [38–43, 53–58] because of its greater chemical shift dispersion and lack of complexities arising from spin–spin coupling and overlap of resonances with those arising from solvents. In contrast to the rather crowded and poorly resolved ^1H NMR spectrum, the proton-noise decoupled ^{13}C NMR spectrum is usually well-resolved and has few overlapping lines, and therefore is inherently easy to interpret. But it is difficult to assign chemical shifts to specific carbon atoms because chemical shift differences among ring carbons are quite small. The ^1H NMR spectrum of the anomeric region may not be straightforward for determining the number of the monosaccharides constituting an oligosaccharide, because acylated methine or methylene in the case of the acylated oligosaccharides and/or hydroxymethine and hydroxy methylene resonances of the non-sugar residue also absorb in the same chemical shift region. Such a situation does not usually hamper the analysis of ^{13}C NMR because anomeric carbon signals resonate in a distinctive region 90–112 ppm (Table 4). This is true for *O*-glycosides only and anomeric resonance in the case of *C*-glycosides, being monooxy-substituted, appear in the chemical shift range 70–80 ppm [41, 57].

The appearance of anomeric resonances in a well-separated chemical shift range of 90–112 ppm helps greatly in determining the number of *O*-linked monosaccharides and in estimating their relative proportions, provided that none of the carbons from the appended group, including the aglycone, absorb in this region. It depends on the fact that all the anomeric resonances are usually non-equivalent and do not usually superimpose on one another. However, identical structural environments may lead to the overlapping of signals; then the approximately equally integrated absorption intensities of the less intense signals can usually be attributed to a monosaccharide residue which can then be correlated with the intense signals to predict the relative proportion of monosaccharide residues. A symmetrical oligosaccharide moiety can also give rise to very simple ^{13}C NMR spectra due to superimposability of its various resonances. Despite the fact that ^{13}C resonances of reducing end monosaccharides absorb at distinct positions, even then one must be very careful in assigning the number of monosaccharide residues in such cases solely from ^{13}C NMR spectra.

The anomeric resonances are of the methine type in aldoses and of the quaternary type in the case of ketoses. The C-1 resonance of a reducing hexose absorbs at ca 5–10 ppm upfield relative to the chemical shift of C-1 of a glycosidic residue. The C-1 of reducing end residues usually appears in the region 90–98 ppm and C-1 of *O*-linked carbohydrates (non-reducing monosaccharides) appears at ca 98–112 ppm; hence the degree of oligomerization can be predicted. The rest of the methine and methylene resonances absorb between 51 and 86 ppm. The appearance of methine resonances between 52 and 57 ppm is generally associated with amino-substituted

carbon signals of an amino sugar residue [40, 59]. Low field absorption in the region 170–176 ppm reflects the existence of a carboxylic group of hexapyranic acids and/or the carbonyl group of acetamido sugars. The presence of an acetamido sugar may further be complemented by the appearance of methyl resonances in the region 21–24 ppm. The spectral region between 57.7 and 64.7 ppm contains signals for all of the unsubstituted hydroxymethylene resonances C-6, whereas methyl resonances of 6-deoxy sugars generally appear in the region 16–19 ppm. Except for glycosylated C-6, which usually absorbs between 66–70 ppm, the rest of the unglycosylated and glycosylated methine resonances appear in the region 66–85 ppm. Aldoses possess one methylene resonance whereas ketoses have two resonances in the region 60–70 ppm. From the number of carbon signals appearing in the above-mentioned chemical shift range, the number of monosaccharide residues can be determined by subtracting those oxygenated carbon signals which belong to nonsugar residues, since naturally occurring monosaccharides are mostly either hexoses or pentoses; therefore, each hexose and pentose unit introduces either six or five resonances, respectively. Accordingly, in a well resolved ^{13}C NMR spectrum, the number of monosaccharide residues can be easily ascertained, in most cases simply by dividing the total number of signals absorbing between 60–85 ppm (excluding those arising from nonsugar residue) either by five or four, or by a combination of both. A hexose monosaccharide gives rise to five resonances, whereas a 6-deoxy, or a 6-carboxy, hexose and a pentose give rise to four resonances in the above-mentioned chemical shift range. When carefully studied, all these data can yield valuable information about the structure of an oligosaccharide.

Most α - and β -pyranoid anomeric forms are indistinguishable by the chemical shifts of C-4 and C-6, but are readily distinguishable by the chemical shifts of C-1, C-2, C-3 and C-5 because these appear at 2–7 ppm higher field in the case of an α -anomer relative to a β -anomer. An exception to this behaviour is the α - and β -anomeric pairs of mannose and rhamnose, where the C-1 chemical shift is uninformative in establishing anomeric configuration. However, the chemical shifts of C-3 and C-5 are of diagnostic importance because these absorb at 1.5–3.0 ppm higher field in the α -anomer relative to the β -anomer. As glycosylation tends to shift the α -carbon to lower field and the β -carbon to a somewhat high field position (section 4.1) careful attention therefore should be paid to the determination of ring size and anomeric configuration on the basis of chemical shifts as they may lead to ambiguity. After this preliminary analysis, individual ^{13}C resonance chemical shifts can be used to identify the types of monosaccharide present in a carbohydrate moiety and their anomeric state (Table 4).

Furanose sugars are characterized by distinctive chemical shifts (Table 5). Generally, signals in the region 80–85 ppm correspond to C-4 aldofuranose and C-5 of ketofuranose. The chemical shifts for the pyranoside and furanoside forms of the same monosaccharide are quite different and therefore, can be used for the determination of ring size. Moreover, $^1J_{\text{CH}}$ values for anomeric carbons range between 168–174 Hz for methyl aldopentofuranosides [60].

By utilizing the foregoing information, it is possible to establish the anomeric configuration of monosaccharide residues. In the case of ambiguity, a reliable criterion for

Table 5. ^{13}C NMR data for methyl glycofuranosides

Sugar	C-1	C-2	C-3	C-4	C-5	C-6
β -D-Glc	110.0	80.6	75.8	82.3	70.7	64.7
α -D-Glc	103.5	77.7	76.6	78.8	70.7	64.2
β -D-Gal	109.6	81.6	78.1	84.4	71.9	63.8
α -D-Gal	103.5	77.8	75.9	82.7	74.1	63.8
β -D-Man	103.6	73.1	71.2	80.7	71.0	64.4
α -D-Man	109.7	77.9	72.5	80.5	70.6	64.5
β -D-All	109.0	75.6	72.7	83.4	73.8	63.9
α -D-All	103.8	72.3	69.9	85.9	72.7	63.5
β -D-Ara	103.2	77.5	75.7	83.1	64.2	—
α -D-Ara	109.3	81.9	77.5	84.9	62.4	—
β -D-Rib	108.5	74.8	71.4	83.5	63.4	—
α -D-Rib	104.2	72.1	70.8	85.5	62.2	—
β -D-Xyl	109.6	80.9	76.0	83.5	62.1	—
α -D-Xyl	103.0	77.7	76.0	79.3	61.5	—
β -D-Lyx	103.2	72.9	70.7	81.9	62.4	—
α -D-Lyx	109.1	77.0	72.0	81.3	61.2	—
β -D-Fru	60.0	104.7	77.7	75.9	82.1	63.6
α -D-Fru	58.7	109.1	81.0	78.2	84.0	62.1
β -L-Sor	57.7	109.9	80.3	77.2	83.4	62.1
α -L-Sor	60.7	104.2	80.0	76.5	78.8	61.6
β -D-Tag	60.7	105.3	73.4	71.7	82.0	61.9
α -D-Tag	58.8	108.7	75.2	71.9	80.6	60.8

determining the anomeric configuration of D-saccharides occurring in the $^4\text{C}_1$ pyranose form is perhaps from one-bond ^{13}C – ^1H couplings ($^1J_{\text{CH}}$), since the difference in coupling between the two anomeric configurations is generally 10 Hz with the higher value for the equatorial ^{13}C – ^1H coupling, i.e. the α -anomer [38, 39, 61]. The mean $^1J_{\text{CH}}$ values for α - and β -anomers are 170 and 160 Hz, respectively. This distinguishing feature is attributable to the fact that the equatorial C–H of the α -anomer is *gauche* to the two lone pair orbitals of the ring oxygen atom, whereas the axial C–H bond of the β -anomer has one *trans* and one *gauche* interaction [62]. Using this method it is possible to distinguish between the anomeric pairs of rhamnose [63] and mannose [64] which are indistinguishable from their ^{13}C NMR chemical shifts of anomeric resonance. This feature, however, does not differentiate between α - and β -anomers of pentofuranosides due to the close resemblance of $^1J_{\text{CH}}$ value, 168–171 Hz for both anomers [65].

Although it might appear that the complete assignment of the ^{13}C NMR spectrum of a carbohydrate residue is easier than that of a ^1H spectrum with its array of overlapping multiplets, a reliable assignment of the carbon spectrum of an oligosaccharide presents some special problems. As a result of the low natural abundance of ^{13}C , there is no simple analogue of the vicinal coupling of protons which provides a rigorous assignment of the resonances of the sugar ring. Usually assignments are made by analogies of the chemical shifts with the assigned resonances of the constituent monosaccharides and to those of simple oligosaccharides following a buildup scheme which incorporates α - and β -glycosidation effects in an empirical manner. This is because the ^{13}C NMR of an oligosaccharide is closely analogous to the sum of the chemical shifts of the monomeric residues from which it is constituted and the number of induced chemical shift changes (glycosylation shifts). For this reason the ob-

served ^{13}C shielding data of oligosaccharides are closely related to the chemical shifts of signals from each monosaccharide. Hence these can be compared to data for monosaccharide residues having the same surroundings in known structures. This method has been found to be very useful and reliable [66–69] leading to the development of computer programmes for structural analysis of oligo- and polysaccharides [70–72]. However, these are not straightforward, particularly for branched oligosaccharides with multiple substitution points, where such an empirical correlation may sometimes lead to errors.

The multiplicity of carbon signals is a valuable aid to spectral assignment. This information can be restored, while retaining the simplicity of the proton-decoupled spectrum using 'attached proton test' (APT) [73], 'distortionless enhancement by polarization transfer' (DEPT) [74] and related 'insensitive nuclei enhanced by polarization transfer' (INEPT) [75] pulse sequences. These involve transfer of polarization from ^1H to ^{13}C and are useful for distinguishing between methine, methylene and methyl carbons, each of which is directly coupled to a different number of protons. Such information is particularly valuable for establishing the structure of appended groups which correspond to the aglycone in glycosides. For carbohydrates, most of whose carbons are of the methine type, such experiments are less useful, serving mainly to provide rigorous identification of C-6 in hexapyranosides. The methyl resonances of 6-deoxy and acetamido sugars absorb at distinct chemical shift ranges (Table 3); thus these can be identified on the basis of their characteristic chemical shifts.

Another method which has greatly added to the solution of ^{13}C assignments in cases of discrepancies is the deuterium isotope shift [76–78], which depends on the difference between the chemical shifts of carbon atoms connected to OH and that after deuterium exchange, i.e. now connected to OD. Deuterium-induced shifts are usually less than 1 ppm [78, 79]; therefore highly accurate chemical shift measurements are required for these experiments. Since the deuterium isotope shift depends on whether a ^{13}C is α or β to a hydroxyl group, it should be possible to distinguish not only the signal assigned to glycosidically-linked carbons, but also those which occupy an adjacent position. Such empirical interpretation requires some caution.

Although the chemical shifts of carbonyl carbon resonances in peracetylated saccharides [80–82], the benzylic methylene carbon resonances of perbenzylated saccharides [83] and methyl carbon resonances of permethylated saccharides [84, 85] have been correlated with the primary structure, reactions do not proceed in a quantitative manner. Moreover, starting material cannot be recovered quantitatively which is important if biological evaluation is required. Therefore, such derivatization should not be carried out routinely but may be of some use in resolving some specific problems.

Two-dimensional NMR spectroscopy

One-dimensional NMR methods yield limited information for the determination of the complete structure and stereochemistry of oligosaccharides including complex saponins. No systematic method for a complete structure analysis has resulted, mainly because of the severe resolution problems encountered. Since most

oligosaccharide proton signals fall within a 2 ppm chemical shift range, substantial overlap of multiplets occurs. These difficulties can be overcome by the use of modern high-field NMR experiments. The critical requirement is the unambiguous assignment of the ^1H resonances of individual sugar residues.

For interpretation of ^1H NMR spectra of oligosaccharides which are not identical to those closely related to known compounds, complete assignment of the methine and methylene resonances in the poorly resolved groups of signals in the region 3.2–4.0 ppm adds greatly to the structural information. Recognition of NMR signals belonging to closed spin systems, i.e. to individual sugar residues, is always the first stage of structural analysis. Since the values of the coupling constants are related to the stereochemistry of the pyranoside ring, it is important to observe the individual components of the multiplets in the crosspeaks for carbohydrates.

Composed mainly of linear chains of coupled spins, carbohydrates are especially suited to spin correlation methods, such as decoupling or two-dimensional shift correlation (COSY) and related techniques, for identification of all the protons present in a given sugar residue. The general approach is to assign an isolated resonance, often an anomeric proton (4.3–5.4 ppm) or the methyl resonance (1.2–1.4 ppm) in 6-deoxy sugars, then to correlate spins in a step-wise manner around the spin system of the ring. To identify these constituent sugars, one also needs numerical values for the vicinal coupling constants of all ring protons. However, spin correlation can be done by one-dimensional difference-decoupling if only a few spin assignments are needed. In most instances, two-dimensional methods are preferred because they are more efficient for the simultaneous determination of a large number of spin correlations.

There are two fundamental types of 2D NMR spectroscopy: *J*-resolved spectroscopy in which one frequency axis contains spin coupling (*J*) and other chemical shift (δ) information, and correlated spectroscopy in which both frequency axes contain chemical shift (δ) information [86–90]. Chemical shift-correlation maps are extremely useful in structural analysis of carbohydrates, providing greatly enhanced resolution of the usually crowded regions of the conventional 1D spectra. Various two-dimensional NMR techniques enable one to identify the components of an oligosaccharide without relying on analogy with any reference data.

Homonuclear-J-resolved spectroscopy (HOMO 2DJ). *J*-resolved spectroscopy [91] is used to resolve overlapping multiplets by giving spectra which have chemical shifts on one axis and scalar coupling on the other. It can provide unprecedented dispersion of the ^1H NMR spectra of carbohydrates [52, 92], but leaves unsolved assignment of individual resonances when strongly coupled nuclei are involved and/or multiplets originating from different spin systems overlap, as frequently occurs in carbohydrates [93]. Since coupling constants are of similar magnitudes for many monosaccharides, measurement of coupling constants alone, therefore does not lead to assignments. The usefulness of the method declines with the increasing number of sugar residues and becomes of limited value in studies of oligosaccharide structure due to overlapping of mutually coupled signals which cause distortions in the multiplet pattern and prevent the use of cross sections to observe individual multiplets and to extract the desired ^1H – ^1H couplings.

Correlated spectroscopy (COSY). Identification of monosaccharide units is first approached by analysing the ^1H homonuclear shift-correlation spectra. The conventional way involves use of COSY which identifies direct J -coupling (e.g. geminal and vicinal) [89, 94, 95]. Therefore, COSY spectra contain information on spin-coupling networks within the constituent residues of the oligosaccharide through the observation of crosspeaks. The multiplet shape is characteristic of pyranosides since the size of the coupling constants is determined by stereochemistry (*trans* or *gauche*) of the protons which are mutually coupled. Assignment of this spectrum by coupling-correlation requires an initial point for identification of the individual spin systems of sugar rings. Since the anomeric proton is connected to a carbon bearing two oxygen atoms, it is generally the most downfield ^1H signal to make it a convenient starting point for the assignment.

Within a typical aldohexopyranosyl ring, the coupling net work is unidirectional, i.e. H-1 couples to H-2, H-2 couples to H-1 and H-3, H-3 couples to H-2 and H-4 and so forth; the absence of coupling, for example, between H-1 and H-3 or H-5 confers this unidirectionality and thus simplifies interpretation. However, the presence of no or small couplings between vicinally-related protons, for example, coupling between H-4 and H-5 ($J_{4,5} = 2\text{--}3\text{ Hz}$) of a galactopyranosyl residue and coupling between H-1 and H-2 in a mannopyranosyl residue, prevents detection of cross peaks and, thus, obviates determination of a complete set of couplings up to H-6 within the ring. However, it would be illusory to expect it to be sufficient for an unequivocal assignment of all resonances of an oligosaccharide. But the overlap of other proton resonances often leads to ambiguities or failures of this approach.

Pure absorption, phase sensitive COSY (PS-COSY). The basis for establishing remote connectivities by PS-COSY [89, 94, 95] lies in the capability of COSY crosspeaks to display the entire coupling information concerning the protons involved. Thus, a crosspeak obtained at frequency F2 (horizontal axis) of a proton and frequency F1 (vertical axis) of another proton not only shows the coupling between themselves (active couplings), but also coupling between these and other protons (passive couplings). Cross-sections through this peak parallel to F2 show the multiplet pattern of this resonance, whereas cross-sections parallel to F1 display the multiplets of another proton. Active coupling appears in anti-phase splitting along both frequency axes, whereas coupling with other vicinal protons (passive coupling) gives rise to an additional in-phase splitting of each of the antiphase components along the appropriate frequency axis. The multiplet components of opposite phase occur as positive and negative signals in crosspeaks. With the above in mind, remote connectivities can easily be traced along the chain of crosspeaks displaying couplings of identical magnitude, with active becoming passive (and vice versa) after passing through the point of degeneracy. The values of $J_{2,3}$, $J_{3,4}$ and $J_{4,5ax}$ are in the range 8–10 Hz for a diaxial *gluco* type-configuration, whereas $J_{4,5}$ is ca 2 Hz in a *galacto* type-configuration. The analysis of the fine structure of crosspeaks is of far greater importance, if identification of the constituent sugar residues and the conformations of their hydroxymethyl groups are of interest.

The fine structure of the H-1/H-2 crosspeak can be used to establish α - and β - anomeric *gluco* or *galacto* and

manno configurations (Fig. 1) [96]. The H-1/H-2 crosspeak shows a ca 3 Hz active coupling and further multiplicity via passive coupling of ca 9 Hz for α -*gluco* and α -*galacto* configurations. In the case of β -anomeric *gluco* and *galacto* configurations, the value of passive coupling remains the same as for the α -configuration but active coupling is ca 7 Hz. Passive coupling is due to H-2/H-3 couplings, whereas active coupling is due to H-2/H-1 couplings in both cases. Measurement of passive coupling in the H-3/H-2 crosspeak can also be employed for the determination of anomeric configuration, because passive coupling to H-1 would be ca 3 Hz in the case of an α -anomeric configuration whereas its value would be ca 7 Hz, for a β -anomeric configuration (Fig. 1). Active, as well as passive couplings, are of ca 2–3 Hz in the case of a *manno* configuration. To distinguish a *gluco* from a *galacto* configuration, their differing $^3J_{3,4}$ values can be read along the F1 axis from the H-2/H-3 crosspeaks where these couplings will be passive. Thus, passive coupling at the H-2/H-3 crosspeak with H-4 will be ca 10 Hz in the *gluco* configuration but is usually not observed in the case of the *galacto* configuration due to the small H-3/H-4 coupling (Fig. 2).

Double-quantum filtered COSY (DQF-COSY). Better visualisation of crosspeaks which are close to diagonal can be achieved by the introduction of a double quantum filter (DQF) [97] which generates a COSY spectrum having both crosspeaks and a diagonal multiplet antiphase structure. This sequence preferentially attenuates the single-quantum resonances of the diagonal with respect to the crosspeaks and also suppresses the detection of the spin-isolated protons, such as those arising from solvent or isolated methyl groups. It provides a clear and accurate way of obtaining chemical shift values coupled protons. It not only provides characteristic multiplicity within the crosspeak, enabling identification of particular sugar units, but also provides semiquantitative information on the coupling constants of protons involved in the crosspeaks.

The analysis of this type of spectrum is straightforward as the direct connectivities between two coupled protons are reflected by the two single-quantum transition crosspeak located at normal chemical shift values on the single-quantum transition axis F2, and also on the double-quantum transition axis F1; these crosspeaks are equidistant from the diagonal. The multiplicities of the crosspeak reflect the coupling pattern of the given resonance as discussed for PS-COSY spectra and shown in Fig. 1.

Strong coupling, which arises when two coupled protons have similar chemical shifts, leads both to some distortion of the expected multiplet shape and to COSY crosspeaks which lie close to diagonal. The former effect interferes with determination of sugar stereochemistry, the latter interfering with tracing the chain of spins within the sugar residue. If multiplet distortion is not too severe it can be accurately interpreted by spin-simulation, but additional experimental methods are needed to complete the tracing of the spin-connectivity. In these cases, spin-relay experiments or isotropic mixing techniques have been shown to be valuable in assignments of oligosaccharides.

Triple-quantum filtered COSY (TQF-COSY). All the spin systems that contain less than three or more mutually coupled spins are eliminated by the use of a triple-quantum filter [98]. One such system in hexopyranosides

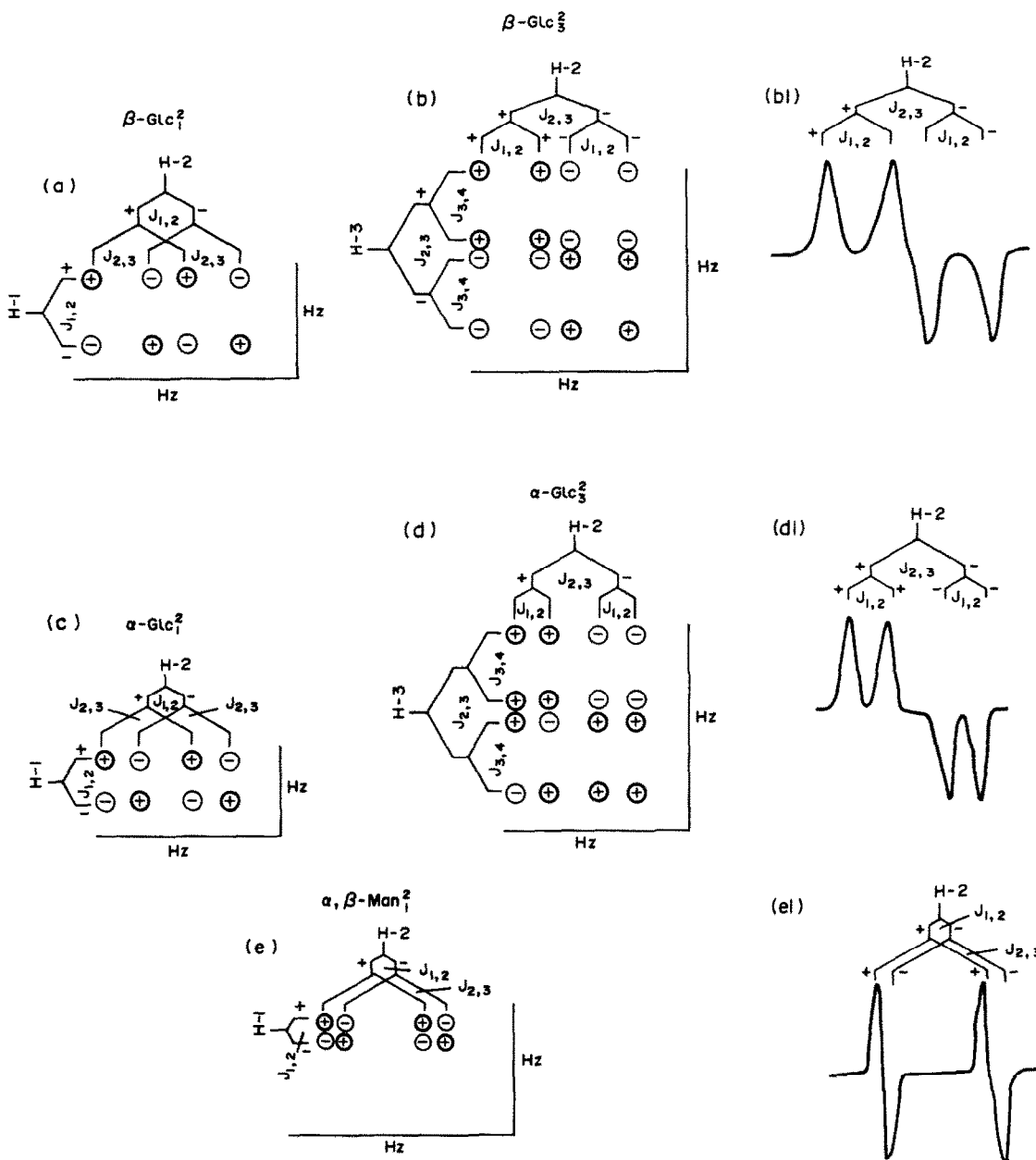


Fig. 1. Diagnostic patterns of crosspeaks (a) H-1/H-2 and (b) H-3/H-2 of β -D-glucopyranose, (c) H-1/H-2 and (d) H-3/H-2 of α -D-glucopyranose and (e) H-1/H-2 of α and β -D-mannopyranose in the phase-sensitive COSY and phase-sensitive DQF-COSY spectrum for determination of anomeric configuration. Positive and negative multiplet components are drawn in thick and thin lines, respectively. (b1, d1, e1) Represent cross-sections of the specific row of b, d and e, respectively. In b1 and d1, the $J_{1,2}$ lead to in-phase splitting, whereas in e1, $J_{1,2}$ lead to anti-phase splitting.

is H-5, H-6 and H-6' which often presents difficulty for assignment if the ring contains equatorial protons with small coupling constants which prevent transfer of coherence from the anomeric proton to H-5 and H-6s in RELAY, TOCSY or HOHAHA spectra. It is worthwhile to mention here that if the two spins in the three mutually coupled proton system are chemically equivalent (i.e. they have same chemical shift) no TQF crosspeak will be seen.

From this it follows, that all crosspeaks will be from conventional COSY, except those correlating the C-5 and C-6 protons will be eliminated in the presence of a triple-quantum filter. Thus, this technique is useful in making assignments of the above-mentioned three or more mutually coupled spins.

Relayed correlation spectroscopy (RELAY). In this approach [99–101], the anomeric proton is not only

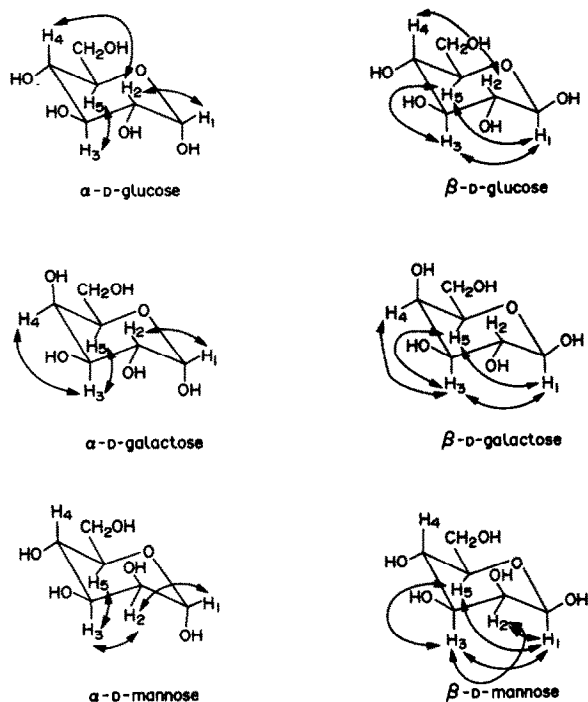


Fig. 2. Possible intra-residue NOE connectivities for various monosaccharides.

correlated with the H-2 proton, but also to other intra-residue protons (H-3, H-4, H-5 and H-6s) in a well-resolved region of the two-dimensional spectrum if the values of the three delays t_1 , t_2 and t_3 in the pulse sequence are chosen correctly [33]. Although crosspeaks between H-1 and H-2, H-3, H-4 and H-5 can be observed, this technique is less successful in the assignment of C-6 protons in view of the conformation dependent values of $J_{5,6}$. The success of RELAY depends on the fact that long-range ($>^3J$) coupling between the pyranosyl ring is effectively zero.

Homonuclear Hartmann-Hahn spectroscopy (HOHAHA). The most useful method of relay of coherence along the chain of spins is the isotropic mixing experiment in which the net magnetization is transferred under spin-locking. This experiment known as HOHAHA [102, 103] is related to total correlation spectroscopy (TOCSY) [104]. From a HOHAHA spectrum, 'J-network' can be determined, where a J-network is defined as a group of protons that are serially linked via 1H - 1H J (scalar) coupling, for example, all the protons of a single saccharide unit belong to the same J-network. A complete spin system can thus be identified if there is at least one resonance in the spin system, such as the anomeric proton, which is well isolated and which has a reasonably large coupling to its neighbouring spin. Therefore, a slice through a HOHAHA spectrum at each anomeric proton along the diagonal yields a 1H subspectrum containing all scalar-coupled protons within that sugar residue. However, the distribution of magnetization around the spin system can be impeded by small coupling, such as typically found between H-4 and H-5 in a galactosyl

residue, which lead to cross peaks up to H-4. To circumvent the bottleneck of small coupling, a one-dimensional version of the two-dimensional HOHAHA pulse sequence can be useful [105].

This experiment is especially useful in sugars for which similar chemical shifts of methine protons leads to many instances of strong coupling or intermediate coupling. In these cases, two correlated signals, close in the spectrum causing COSY crosspeaks, close to the diagonal and are therefore impossible to detect. But since the 2D HOHAHA spectrum contains crosspeak-isolated resonances such as H-1 and the other resonances in the spin system, the relevant peak will be well-resolved from the diagonal. This procedure appears to be adequate for confirmation of known structures, but hardly applicable to totally new structures, since chemical shifts may be very different, owing to the glycosylation-induced shifts occurring in oligosaccharides and the choice of a suitable set of coupling constants would rely on guesswork or studies of related oligosaccharides.

To avoid ambiguities of this sort, it is desirable that assignment procedures should be complemented by the analyses of DQF COSY or PS COSY spectra. A complete assignment of the resonances of each spin system to individual pyranoside rings can usually be achieved by the DQF COSY method, augmented perhaps by HOHAHA, where other methods have been found useful for the solution of specific problems in the assignment.

Nuclear Overhauser effect spectroscopy (NOESY). Proton nuclear Overhauser enhancement (NOE) which depends on proton proximity, can be a valuable assignment aid and in the assessment of molecular conformation (i.e. 3D structure). Such spectra may be conveniently measured by the use of two-dimensional NOESY [106]. This procedure is never circular, since intraresidue NOEs are the principal assignment tool, whereas interresidue NOEs are primarily used for determination of the sequence of sugar residues and also in determining their linkage positions. Crosspeaks are observed in 2D NOESY spectra between proton pairs that are close in space (i.e. typically less than 5 Å). In general, 1,3-diaxial and eq-ax proton pairs in pyranosyl rings produce intra NOESY crosspeaks, i.e. for β -glycopyranosyl residue crosspeaks are observed between H-1 and H-3 (and H-5) whereas a strong crosspeak is observed between H-1 and H-2 in an α -glucopyranosyl configuration. In this way observation of NOE crosspeaks also discriminates between the α - and β -anomers of mannose and rhamnose, for example, strong NOE between H-1 and H-2 will be observed for α -D-mannose whereas NOE from the anomeric proton to H-2, H-3 and H-5 or vice versa, can be observed for β -D-mannose (Fig. 2). The magnitude of a NOE depends not only on the 1H - 1H internuclear distance but also on the rotational correlation time.

INADEQUATE spectroscopy (INADEQUATE). The 2D 'incredible natural abundance double quantum transfer experiment' provides direct information on carbon-bonding and, therefore, can be used to trace the entire carbon skeleton of the molecule [107]. The presence of a pair of double quantum peaks normally indicates presence of a bond; however, these may be absent from carbon atoms with long relaxation time, strongly coupled carbons and if the chemical shift difference is large. In spite of its exceptional value, this technique could not be employed routinely to oligosaccharides, because of its low sensitivity.

Heteronuclear 2D-NMR spectroscopy

Sometimes the proton resonances of an oligosaccharide are too overlapping to be disentangled by homonuclear correlation alone. In such cases heteronuclear correlation maps may enable the assignment of ^1H resonances, because in such a spectrum one observes connectivities between ^1H and ^{13}C chemical shifts. This method spreads the ^1H NMR spectrum in the ^{13}C dimension, thus greatly improving the resolution and eliminating the effects of strong ^1H -couplings. Usually ^1H - ^{13}C cross-peaks do not superimpose until the ^1H and ^{13}C chemical shifts are identical due to the presence of a very similar chemical environment.

After establishing ^1H NMR assignments by the above-mentioned methods, it is possible to identify the monosaccharide units that correspond to a particular J -network by a one-bond heteronuclear (^1H - ^{13}C) correlation spectrum. The group of ^{13}C resonances that correlate will all be members of a J -network consequently will represent one monosaccharide unit. By comparison of this group of resonances with known assignments of monosaccharide and model compounds, it is a straightforward matter to identify the monosaccharide, their furanose and pyranose forms, and to establish the anomeric configuration of the sugar. These correlation maps shows that the more strongly shielded anomeric proton, i.e. H-1 of an α -anomer, is appended to a less strongly shielded ^{13}C nucleus, thus, revealing the appearance of an anomeric carbon resonance of an α -anomer at higher field than that of the corresponding β -anomer. It must, however, be emphasized that such methods can be used independently in order to establish the structure of a monosaccharide if reference data do not exist or if data are not totally consistent due to sample conditions.

^{13}C - ^1H Heteronuclear Correlated Spectroscopy (HETCOR). In such a spectrum, each cross peak arises from connectivity between a ^{13}C nucleus and its directly-bonded proton having the coordinates (^1H , ^{13}C). It must be mentioned that in such experiments only 1% of the protons which are coupled to ^{13}C are actually detected, so that the much stronger signal of 99% of protons attached to ^{12}C must be suppressed. In ^{13}C - ^1H correlations, experiments are carried out in the ^{13}C detected mode [108] and one detects ^{13}C signals during t_2 and measures along the f_1 axis the spectrum of the protons attached to each carbon. [109]. Chemical shifts of protons attached to each carbon can be deduced from the greater resolving power of the ^{13}C spectrum, but the low sensitivity of the method presents a major problem. Two-dimensional heteronuclear correlation via long-range coupling (COLOC) has been found to be useful in determining the connectivity of sugar to aglycone [110]. Despite the fact that HETCOR techniques have been frequently employed in the structural analysis of carbohydrates [52, 111, 112], recently it has been almost entirely replaced by two-dimensional ^1H -detected (^1H , ^{13}C) chemical shift correlations.

^1H Detected ^1H - ^{13}C chemical shift correlation spectroscopy. These experiments are analogous to HETCOR but instead of observing ^{13}C , the more abundant ^1H is detected which leads to improvements in sensitivity sufficient enough to make ^{13}C - ^1H correlation spectroscopy of oligosaccharides a real possibility. Such experiments enable one to trace scalar connectivities between ^1H and ^{13}C atoms through indirect detection of the low natural

abundance nuclei, ^{13}C via ^1H nuclei. Experiments designed to date fall into two categories: (i) heteronuclear multiple-quantum coherence (HMQC) [113] and heteronuclear single quantum coherence (HSQC) [114]. A modified version of HSQC has been recently proposed which allows both one-bond and multiple-bond (^1H , ^{13}C) correlation spectra to be obtained with high resolution in the ^{13}C domain [115].

There are several variants of inverse (reverse) detection, but all rely on the generation of multiple quantum coherence between ^1H and ^{13}C energy levels. These provide correlations between directly-bonded ^1H and ^{13}C resonances if ^{13}C decoupling is used. This further extends its power in deriving complete assignment of the carbon spectrum from the assigned proton spectrum, or vice versa, and it has been employed for the establishment of the structures of triterpenoid glycosides and oligosaccharides [116, 117].

However, if ^{13}C decoupling is not used during acquisition, then correlation peaks in the 2D spectrum appear minimally as doublets in the ^1H dimension [117]. The separation between signals is equal to a one-bond ^1H - ^{13}C coupling constant and this coupling constant is useful in determining the anomeric stereochemistry of the monosaccharide unit. The ^1H - ^1H coupling is also retained, hence, a doublet in the 1D ^1H spectrum will appear as doublet ($J_{\text{H,C}}$) of doublets ($J_{\text{H,H}}$) in the 2D spectrum.

Heteronuclear 2D J-resolved spectroscopy. The value of one-bond ^{13}C - ^1H coupling constants (J_{CH}) can be measured by this technique; they are important parameters for the establishment of anomeric configuration [38, 39]. By the use of 2D DEPT or POMMIE (phase oscillation to maximize editing) one can get $J(\text{CH})$ spectral editing [118].

Hybrid methods. These methods provide a powerful alternative for obtaining further connectivity information, especially when the ^1H spectrum is highly congested. Such techniques usually utilize the resolving power of the heteronuclear experiment and coherence transfer to some neighbouring protons either by COSY, relay/TOCSY (HOHAHA) and NOESY.

HMQC-COSY. Using this technique [119] one observes crosspeaks at each ^{13}C frequency (F_1 dimension) which relate a carbon atom to the ^1H resonance of a proton connected by one bond which is split by a large J_{CH} coupling and exhibits a relay peak to the vicinal proton resonances which are not split by CH coupling. Since direct peaks are split by large one-bond ^1H - ^{13}C couplings, relay peaks that appear in the middle of the split can be accurately assigned. Large vicinal proton coupling gives stronger relay peaks whereas smaller couplings tend to give weak peaks. Thus, this combination method is a powerful method for the assignment of strongly-coupled vicinal ^1H signals [120].

HMQC-RELAY/TOCSY (HOHAHA). Instead of observing direct responses between one bond bonded ^1H and ^{13}C nuclei, one observes with this technique, connectivity of a ^{13}C signal to the relay peak, usually an adjacent proton [121]. The response is dependant upon the duration of the isotropic mixing interval as one observes only a one bond ^1H - ^{13}C correlation with short mixing time, whereas responses to adjacent protons can be observed by increasing the duration of mixing [122].

HMQC-NOESY. This method [123] allows the effective development of proton-proton NOEs in addition to

direct ^{13}C - ^1H correlation peaks. Thus, at a particular ^{13}C frequency one observes crosspeaks to those ^1H resonances to which NOE of that directly bonded ^1H resonance has been transferred, along with crosspeaks for directly bonded ^1H nuclei. Thus, this method could be of value for proton assignments where COSY and NOESY fail when key proton signals are poorly resolved because of the proximity of the correlating off-diagonal response to the diagonal. This technique has been employed for the determination of stereochemical assignments of a carboxylic nucleoside [124].

The application of some of the above-mentioned techniques is now demonstrated by taking 2,3-dideoxy-2,3-diacetylamidoglucopyranose as an example. The sample was not exchanged with D_2O for the purpose of measurement of chemical shifts of exchangeable amide (NH) and hydroxyl (OH) protons. The ^1H NMR and broad-band decoupled ^{13}C NMR spectra are shown in Fig. 3. The ^1H NMR displays amide resonances in the region 7.98–7.51 ppm, acetyl methyl singlets at 2.01 and 1.96 ppm, while the rest of the ring protons and hydroxylic protons appear in the region 6.94–3.49 ppm. In the anomeric region, there are signals at 5.28–5.30 ppm and 4.79–4.84 ppm, but none of the signals can be identified from anomeric resonances which also show typical doublet splitting due to their coupling with the geminal hydroxyl proton.

The broad-band decoupled ^{13}C NMR spectrum, however, is more informative and shows two anomeric resonances at 96.8 and 90.6 ppm and four amido-substituted carbon signals at 56.4, 55.8, 53.5 and 51.9 ppm, thus inferring the presence of both of the anomeric forms (α

and β). Careful examination of the spectrum led to the identification of two sets of signals with different intensities. Except for the signal at 68.8 ppm, which seems to be common for both anomers, the signals at 90.6, 53.5, 51.9, 73.0 and 61.5 ppm were intense, whereas signals at 96.8, 55.8, 56.4, 78.5 and 61.7 ppm were weak. Since C-1 of the α -anomer, as in most pyranoses, appears at 4–8 ppm higher field position relative to the β -anomer, the former set of signals could therefore be assigned to the α -anomer, the latter to the β -anomer; assignment to individual ^{13}C resonance, however, is not possible. The value of the one-bond ^{13}C - ^1H coupling constants, 166.7 Hz and 158.6 Hz for the signals at 90.6 and 96.8 ppm are in accordance with the anomeric configuration proposed above. The signals at 61.5 and 61.7 ppm correspond to C-6 from their characteristic chemical shifts and their splitting as a triplet in the proton-coupled ^{13}C NMR spectrum. The signals at 73.0 and 78.5 ppm could be tentatively assigned to C-5 of the α - and β -anomers respectively. The assignment of C-2 and C-3 was not straightforward but this was achieved by rigorous analysis of DQF-COSY and HO-HAHA spectra which resulted in the assignments of ^1H resonances; these were correlated with ^{13}C resonances in a HMQC experiment as discussed below.

Identification of ring protons of the individual anomer is achieved by analysis of the DQF-COSY spectrum using amide ^1H signals (NHs) as the starting point. All the crosspeaks corresponding to the α -anomer could be seen at a higher contour level (Fig. 4). For instance, the amide resonance at 7.51 ppm exhibited a crosspeak at 4.05 ppm which was further correlated with the resonances at 5.28 and 4.36 ppm. The resonance at 5.28 ppm exhibited

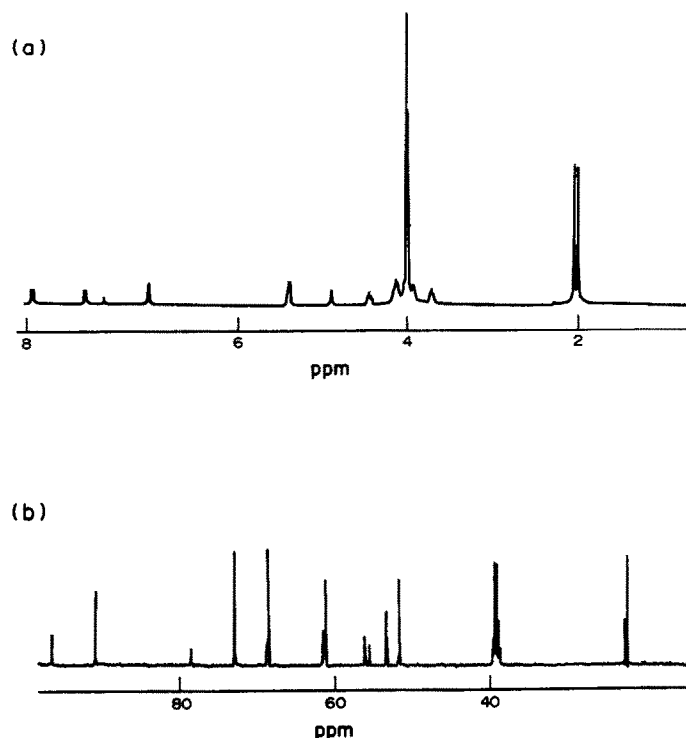


Fig. 3. One-dimensional NMR spectral data for 2,3-dideoxy-2,3-diacetylamidoglucopyranose in benzene- d_6 + $\text{DMSO}-d_6$ (4:1). (a) ^1H NMR spectrum and (b) broad band-decoupled ^{13}C NMR spectrum.

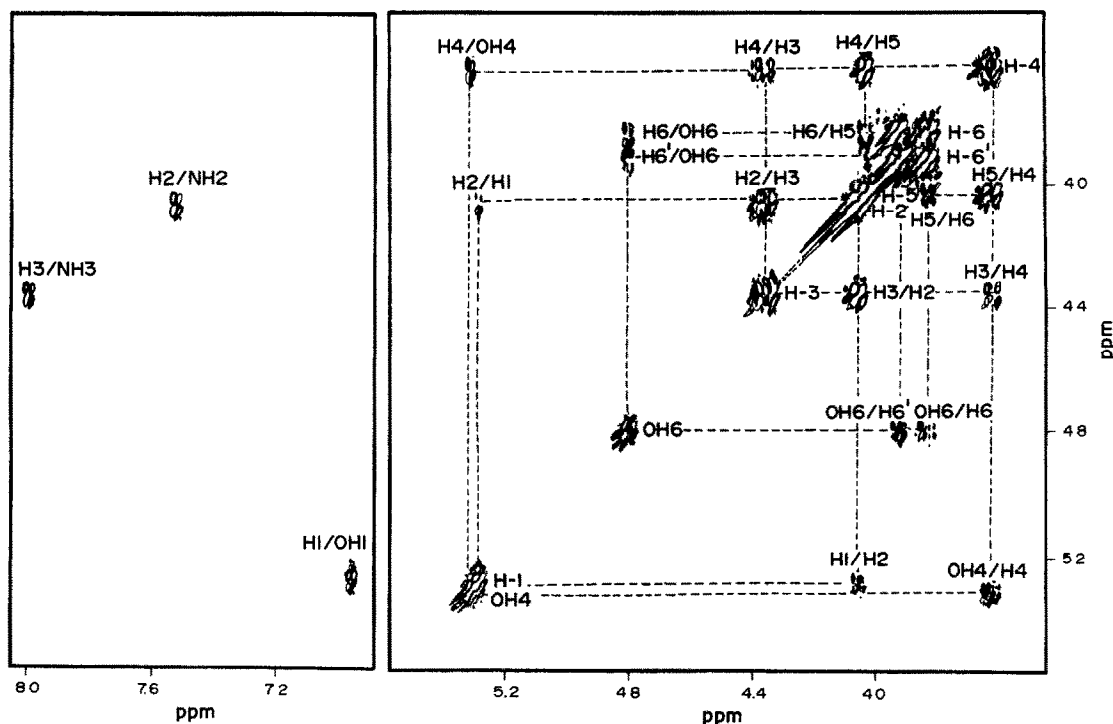


Fig. 4. Diagrammatic representation of through-bond connectivities in the DQF-COSY of 2,3-dideoxy-2,3-diacetylamidoglucopyranose at higher contour level showing crosspeaks for the α -anomer.

crosspeaks at 4.05 and 6.94 ppm. The resonance at 6.94 ppm corresponds to a hydroxylic proton (OH-1) since it shows a crosspeak only to the resonance at 5.28 ppm; accordingly resonances at 5.28 and 4.05 ppm are assigned to H-1 and H-2, respectively. The H-1/H-2 crosspeak shows a small active coupling ($J=3.2$ Hz) and larger passive coupling ($J=10$ Hz) and this residue, therefore, corresponds to the α -anomer. The H-2 resonance at 4.05 ppm, in addition to its crosspeak connectivity to H-1 (5.28 ppm) and NH-2 (7.51 ppm), shows crosspeaks at 4.36, 3.84 and 3.62 ppm due to strong coupling of H-2 and H-5. However, H-3 can be identified at 4.36 ppm since the amide resonance at 7.98 ppm shows a crosspeak to this resonance only. Thus, the resonance at 3.62 ppm corresponds to H-4 from the crosspeak connectivities to H-3 at 4.36 ppm and the hydroxyl proton at 5.30 ppm (OH-4). Both of the H-6 at 3.84 and 3.93 ppm show their correlation with H-5 at 4.05 and the hydroxylic proton (OH-6) at 4.79 ppm.

Analyses of the 2D HOHAHA spectrum of the compared (Fig. 5) were consistent with the above proposed assignments. For each diagonal peak in the HOHAHA spectrum, the scalar-coupled resonances in the spin system could be obtained by examining the cross-section along either the F1 or F2 axis. The anomeric resonance at 5.28 ppm showed crosspeaks up to H-5 along with weak crosspeaks to H-6s.

The ^{13}C - ^1H one-bond correlation through a ^{13}C -decoupled ^1H detected heteronuclear multiple-quantum coherence (^1H [^{13}C]HMQC) spectrum at a higher contour level (Fig. 6) led to the assignment of all ^{13}C resonances of the α -anomer. Correlation of H-2 and H-3 resonances at 4.05 and 4.36 ppm with ^{13}C resonances at

53.5 and 51.9 ppm led to the assignment of these amide-substituted carbons to C-2 and C-3, respectively.

Likewise, the ^1H signal assignments for the β -anomer were traced out from the DQF-COSY and HOHAHA spectra in an analogous manner to that discussed for the α -anomer. The crosspeak connectivities for both anomers and the cross-section taken along the chemical shift of the anomeric resonances and the OH-6s are shown in Fig. 5. Thus, ^{13}C resonance assignment could be obtained directly from the ^{13}C - ^1H one-bond correlated HMQC spectrum at a lower contour level (Fig. 6). The ^1H and ^{13}C NMR chemical shifts of the α - and β -anomers of 2,3-dideoxy-2,3-diacetylamidoglucose are given in Table 6.

Using the present example, the usefulness of DQF-COSY, HOHAHA and HMQC has been demonstrated for the anomers of a monosaccharide but the general strategies for identifying monosaccharide residues of a oligosaccharide, or a glycoside, will be very similar. Once the unambiguous ^1H and ^{13}C resonance assignments for individual monosaccharides have been obtained, they can be combined with glycosylation-induced shifts and *ab initio* methods to determine the site of interglycosidic or sugar-aglycone linkage as discussed below.

DETERMINATION OF INTERGLYCOSIDIC LINKAGES AND OLIGOSACCHARIDE SEQUENCES

Previously, the most common procedure to get such information was by analysis of fragments obtained by chemical or enzymatic degradation. This method has now been replaced by NMR spectroscopy. Once each sugar residue has been identified and its anomeric configuration determined using a combination of above-defined meth-

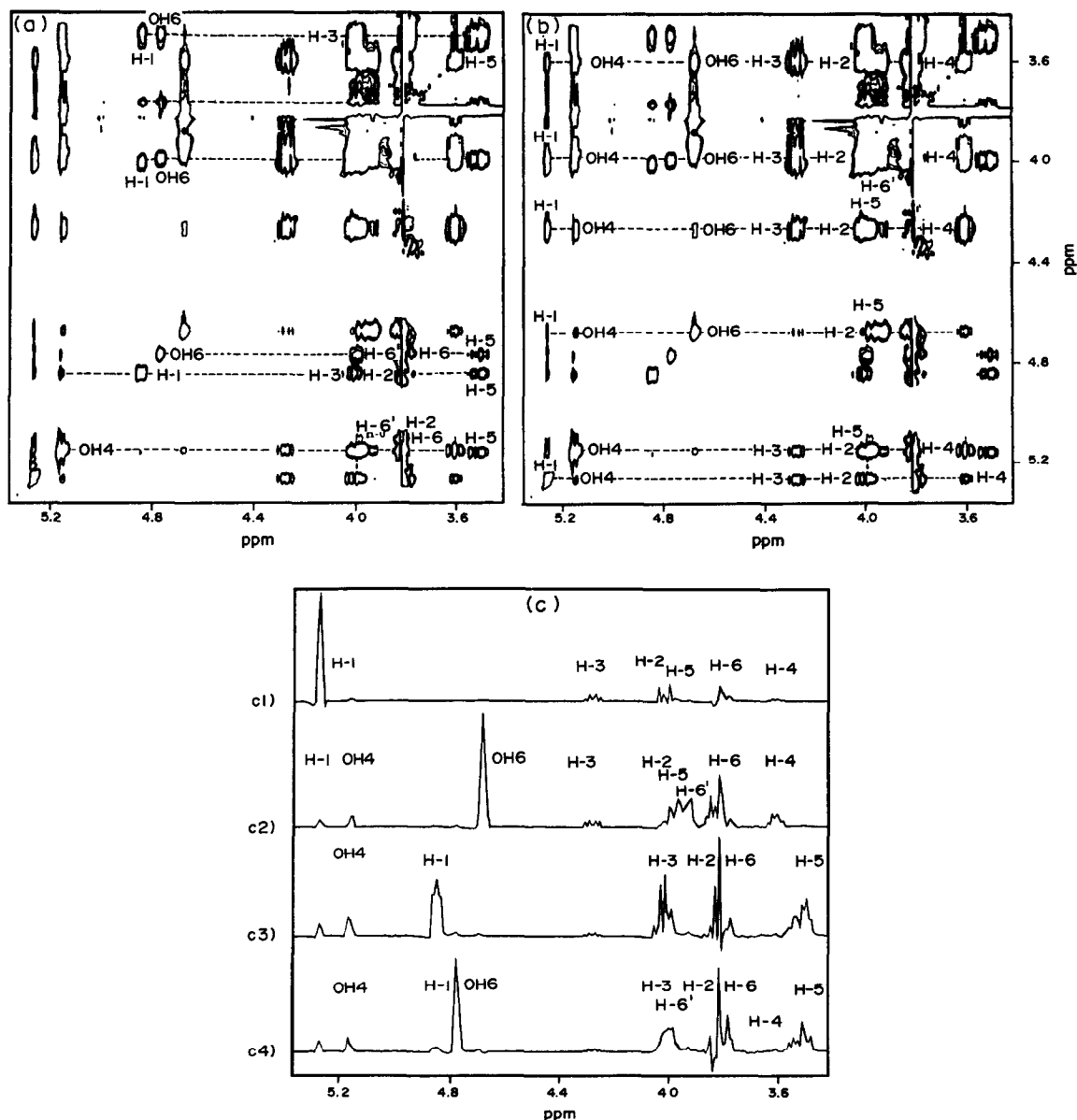


Fig. 5. Phase-sensitive homonuclear Hartmann-Hahn spectrum of 2,3-dideoxy-2,3-diacetylamidoglucose showing crosspeak connectivities (a) for the β -anomer, (b) for the α -anomer and (c) cross-sections along f_1 through the chemical shift of (c1) H-1 (5.28 ppm) and (c2) OH-6 (4.67 ppm) of the α anomer and (c3) H-1 (4.84 ppm) and (c4) OH-6 (4.76 ppm) of the β -anomer.

ods, all that is required to complete the structure determination is to identify the glycosidic linkages via $^3J_{\text{COCH}}$ and their sequence. Vicinal proton coupling, which is very useful in proton assignments for individual pyranoside rings, however, is less valuable for correlating monosaccharide residues to each other. Since, the protons across the glycosidic linkage are four bonds apart, they do not show scalar coupling, and thus no correlation between individual spin systems can be observed by COSY or HOHAHA. Accordingly no sequence information can be obtained from such experiments. Two approaches can be used, depending upon whether prior knowledge of any

structural detail is available or not. The 'glycosylation-induced shift' method is quick and easy to apply if the oligosaccharide is related to any previously examined by NMR, whereas the *ab initio* method in principle requires no prior knowledge of related structures.

Glycosylation shifts (GS)

This method depends on the fact that substitution at a sugar ring by another sugar unit induces chemical shift changes in both ^1H and ^{13}C NMR spectra, referred to as glycosylation shifts, GS.

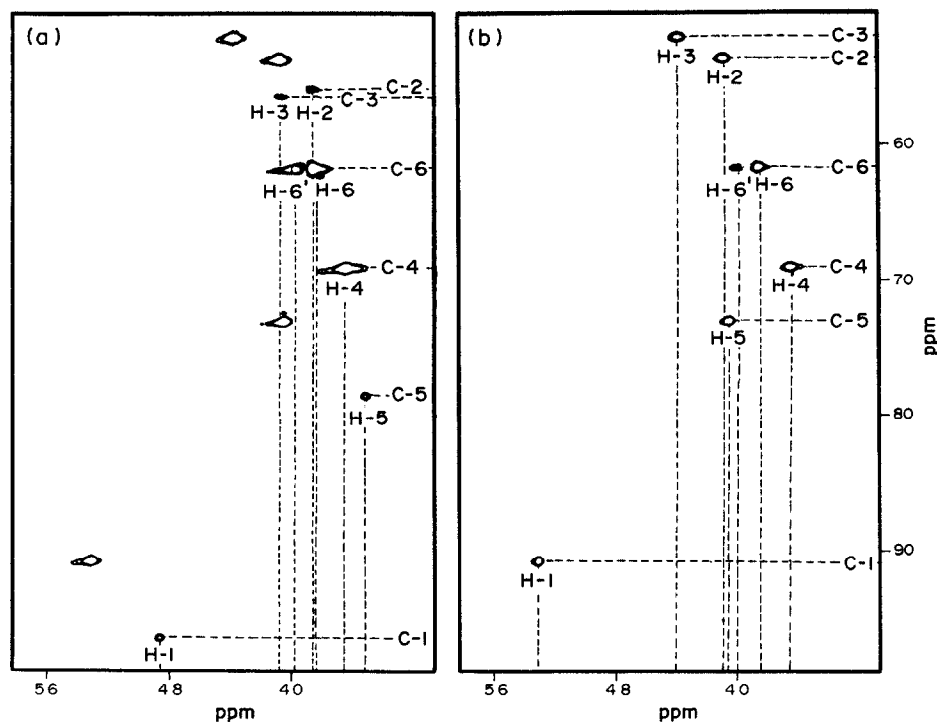


Fig. 6. Diagrammatic representation of ^{13}C – ^1H one-bond correlation through a ^{13}C -decoupled ^1H detected heteronuclear multiple quantum coherence (^1H [^{13}C]) HMQC spectra of 2,3-diacetylaminoglucose (a) at lower contour level crosspeaks for both anomers could be seen, however, ^{13}C – ^1H connectivity is only shown for the β -anomer and (b) at higher contour level which showed crosspeaks for the α -anomer only.

Table 6. ^1H and ^{13}C NMR spectral data for α - and β -2,3-dideoxy-2,3-diacetylaminoglucopyranose*

Atom no.	^1H	α ^{13}C	β ^1H	β ^{13}C		^1H	α ^{13}C	β ^1H	β ^{13}C
1	5.28	90.6	4.84	96.8	2NHCO	—	171.4	—	171.2
2	4.05	53.5	3.86	55.8	3NHCO	—	172.4	—	171.7
3	4.36	51.9	4.05	56.4	3Me	2.01	23.4	2.01	23.4
4	3.62	68.8	3.62	68.8	2Me	1.96	23.8	1.96	23.8
5	4.06	73.0	3.49	78.5	NH2	7.51	—	7.976	—
6	3.84	61.5	3.84	61.7	NH3	7.98	—	7.974	—
	3.93	—	3.93	—	OH1	6.94	—	6.95	—
					OH4	5.30	—	5.30	—
					OH6	4.67	—	4.76	—

*Measured in benzene d_6 –DMSO- d_6 (4:1) at 25°.

In ^1H NMR spectra, glycosylation tends to shift the protons of the glycosylated residue, particularly those at the linkage site, by -0.20 to 0.26 ppm, and those vicinal to it, to lower field (0.03 – 0.31 ppm), whereas other resonances are much less affected. Although one or both of the vicinal protons may be even more affected than at the glycosylation site, the latter can nevertheless be unambiguously determined from the signal for the proton of these three GS-exhibiting protons. Therefore, this method can be applied to those oligosaccharides whose unambiguous ^1H assignments are already available; such shifts have been correlated with different types of inter-

actions around the glycosidic bond [125–128]. Because unambiguous ^1H NMR assignments for oligosaccharides are not often reported, it is not possible at the present time to draw up well-defined rules for GS but it is hoped that these will be soon become available.

An alternative approach utilizes the peracetylation of the oligosaccharide followed by measurement of the ^1H NMR spectrum of peracetylated oligosaccharides, since acetylation causes strong deshielding of the order of 0.4 – 1.5 ppm with little effect on neighbouring protons including the glycosidic position, i.e. the ^1H resonance at the site of a glycosidic linkage absorbs almost at the same

position as its peracetylated derivative as well as that of the underivatized oligosaccharide; therefore, determination of the position of glycosylation is straightforward.

Glycosylation shifts in ^{13}C NMR are relatively regular. The glycosylated carbon shifts to lower field by 4–10 ppm (the α -effect). The resonances of carbon atoms adjacent to the linked carbon atom are usually shifted upfield by a small amount (+0.9 to –4.6 ppm), but not necessarily similar amounts (β, β' -effect), whereas other carbon resonances remain virtually unaffected. These glycosylation effects depend on the configuration at the anomeric centre of the glycosidating pyranose and the absolute configuration of both pyranose residues. These effects have been correlated with spatial proton–proton interactions, which cause polarizations of the C–H bond. The 1,3 interactions between H-1' of the glycone and H- α of the aglycone causes a downfield shift of the corresponding C-1' and C- α , and 1,4-interactions between H-1' and H- β cause an upfield shift. β, β' -Effects have been reported to show a relationship with the conformation of glycosides [125–131]. The correlation between ^{13}C chemical shifts for both glycone and aglycone carbons with one of the torsional angles (ψ) can be used for the determination of the conformation of the glycosidic linkage.

Glycosylation effects in the disaccharide fragments of unbranched oligosaccharides and polysaccharides are almost identical to those in corresponding disaccharides and hence they are transferable parameters. This similarity enables the evaluation by an additive scheme from the ^{13}C NMR spectra of carbohydrates of known structures starting from chemical shift data for the constituent monosaccharides and the average values of the glycosylation effects. This leads to establishment of the primary structure of carbohydrates on the basis of their ^{13}C NMR data. Therefore, the glycosylation site can be determined by comparison of the chemical shifts of oligosaccharide with their respective methyl glycosides. The establishment of regularities in the effects of glycosylation allowed the development of a computer-assisted approach to the structural analysis of linear regular polysaccharides on the basis of ^{13}C NMR data [66–68, 129–131].

Additivity for both ^1H and ^{13}C glycosylation-induced shifts, in general, holds good with small deviations attributed to the effect of conformation and concentration for di-, tri- and linear regular polysaccharides. In spite of the additive nature of glycosylation-induced shifts, they become less reliable, particularly for an oligosaccharide with a monosaccharide at the branching point substituted at any vicinal hydroxyl groups. In such cases, glycosylation effects may differ considerably from those in the corresponding disaccharide fragments. Interresidue NOE and long-range ^{13}C – ^1H correlated spectroscopy must then serve as a basis for determination of the glycosylation site.

Ab initio methods

This term has been used for methods by which the sequence of an oligosaccharide is determined without the prior knowledge of the type of structure, in the sense that no compositional data were available.

Spin lattice relaxation time. The normalized or average spin lattice relaxation time (NT_1) for sugar carbons in glycosides increases with the increase of distance from the aglycone [132] due to segmental motion. Thus, terminal

sugars exhibit a higher NT_1 than NT_1 for the inner sugars. This technique has been applied for sequencing monosaccharides from several glycosides [133–136].

NOE difference spectroscopy. Irradiation of specific protons for example, those anomeric protons in steady-state NOE give the negative NOE to the aglycone proton in NOE difference spectroscopy, hence, difference NOE (DIFNOE) provides information about the interglycosidic linkage and similarly the position of the appended groups can also be established. This technique has been widely applied in the structural establishment of anthocyanins [137–143], related flavonoid C-glycosides [144–146] and a steroid glycoside [147].

ROE difference spectroscopy. Rotating-frame Overhauser effect (ROE) difference spectroscopy (ROEDS) has also been used to identify glycosidic linkages. Intensities of NOE signals indicate the linkage site, as a greater NOE is usually noted for the proton at the linkage site as compared to the other protons of the aglycone sugar residue upon irradiation of the anomeric proton of the glycosylated residue. This technique has been used to establish the structure of a triterpenoid bisdesmosidic hexaglycoside [148].

Long-range selective proton decoupling. In aromatic compounds, long-range selective proton decoupling (LSPD) [149] has been found to be quite valuable in structural elucidation. Here selective irradiation of a proton showing long-range coupling to an aryl carbon under gated-decoupling, simplifies its multiplicity pattern, thus identifying its long-range coupling between an irradiated proton and a particular carbon atom. In a similar way, irradiation of an anomeric proton changes splitting for that carbon to which it is glycosidically linked, hence identifying the site of glycosidation [150].

1D-Selective INEPT. This method [151, 152] relies on long-range ^1H – ^{13}C scalar coupling interactions ranging between 3 and 10 Hz and can be used to establish connectivity between an anomeric proton and the carbon atom, three bonds away from the aglycone. Selective enhancement of this carbon is usually observed, i.e. irradiation of the anomeric proton will lead to the appearance of the aglycone carbon to which it is glycosidically linked and in a similar way irradiation of the aglycone proton will lead to the appearance of the anomeric carbon of the glycone residue. The glycosidic linkage in the case of flavanoid glycosides [153, 154], a cycloartane glycoside [155, 156] and oligosaccharides [157, 158] has been identified by the application of this technique.

^1H – ^{13}C nuclear Overhauser effect spectroscopy. Selective saturation of the anomeric proton of a sugar unit causes an overall negative enhancement of the aglycone carbon in the NOE difference ^{13}C NMR spectrum. Employing such a method, glycosidic linkages in oligosaccharides have been identified [159].

Long-range and delayed COSY. The coupling between an anomeric and an aglyconic proton is usually very small even using resolution-enhanced spectra, but it can be detected by long-range COSY [160] and delayed COSY [161] since these allow the observation of 4J intersugar couplings between the anomeric proton and aglycone protons. This method has been employed for the sequencing of sugars of triterpenoid saponins [161, 162] and oligosaccharides [163].

Nuclear Overhauser effect spectroscopy (NOESY). The presence of an interresidue NOE from the anomeric

proton of a particular sugar residue to proton(s) of the other sugar residue in the case of oligosaccharide, or to non sugar residues in the case of glycosides, defines the glycosidic linkage between the two residues. NOE connectivities are most often observed between the anomeric proton and the proton connected to the carbon atom of the linkage (the aglycone proton) (Fig. 5). This method has been found to be of wide applicability for the determination of the structures of a number of naturally occurring glycosides [164–174]. The effect depends upon the local conformation about the glycosidic linkage. Therefore, some care must be used in deducing the nature of the interglycosidic linkage from proton NOE data, because NOE depends not only on the proximities of protons but also on the correlation time of the molecule. Since NOE also depends on the distances between protons, it is possible, in principle, to determine interproton distances directly from NOE data. Thus, it is a major source of experimental information for determining conformation of carbohydrates.

NOE in rotating frame (ROESY). Due to the well-known problems involved with NOE measurements at medium field strength for medium-sized molecules, a 2D NOE in a rotating frame (ROESY) [175–177] can be of importance. In cases where attempts to obtain reliable NOE crosspeaks are unsuccessful, a ROESY spectrum can show all NOE crosspeaks defining interglycosidic linkage. Because NOE is a function of molecular rotational time, which itself depends on the size and shape of the molecule, viscosity of the medium and temperature are important. Using this method it is possible to distinguish between direct and 'three spin' NOEs from their opposite signs.

The previously proposed structures of the triterpenoid glycosides chrysantellins A [178] and B [179] have been recently revised based upon ROESY analyses, combined with other NMR techniques [180, 181]. The application of this technique has led to the determination of the structures of triterpenoid saponins [182, 183], carotenoid glycosides [184], flavonoid glycosides [185] and oligosaccharides [186].

Long-range C–H J-resolved 2D NMR spectroscopy. Selective irradiation of the anomeric proton using this technique [187] causes a split of the aglycone carbon but it has been rarely employed for the structural establishment of glycosides and oligosaccharides. To the best of our knowledge there is only one instance of a flavanone glycoside [188] in which the position of the glycosidic linkage was established utilizing this methodology.

Long-range heteronuclear chemical shift correlation. Among the various techniques [189] available for correlating ^1H NMR chemical shifts to ^{13}C NMR chemical shifts to which it is long-range correlated, the most popular are COLOC (correlation by long-range coupling) [190], XCORFE (X-nucleus correlation with fixed evolution time) [191] and HMBC (heteronuclear multiple bond correlation) [192]. Once the carbon spectrum has been completely assigned, an unambiguous determination of the glycosidic linkage can be obtained from the long-range C–H correlation. The aglycone proton exhibits its correlation with the glycone carbon and conversely the glycone proton exhibits its correlation with the aglycone carbon.

Because HMBC is ^1H -detected, it is a very sensitive method for establishing glycosidic linkages. This method, in addition to the intraresidue multiple bond correlation,

valuable for confirming ^{13}C and/or ^1H assignments, provides interresidue multiple bond correlations between either the anomeric carbon and the aglycone proton or the anomeric proton and the aglycone carbon, and thus serve to identify interglycosidic linkage (Fig. 5). The correlation peak appearing at the chemical shift of the ^1H anomeric resonance establishes the position of the interglycosidic linkage. A similar correlation peak between the ^1H chemical shift of the aglycone and the C-1 anomeric of the glycone can also be observed. However, the intensities of the correlation peaks depend on the multiplicity of the respective proton resonance, i.e. broad multiplicity of proton resonances leads to poor crosspeaks. Although these coupling constants are known to depend on geometry, unambiguous determination of the linkage requires only that either one of these couplings be greater than *ca* 5 Hz, permitting detection in long-range ^{13}C – ^1H correlation spectra. Discrimination between intra- and intersaccharide correlation is made by reference to the established ^1H resonance assignments and by an iterative comparison of the one- and multiple-bond correlation maps.

This technique has been applied to the determination of glycosidic linkages in flavonoid *O*-glycosides [193–196], hydrolysable tannins [197, 198], a steroidal saponin [199], a phenyl propanoid glycoside [200] and other natural products [119, 201–203]. An additional advantage to those long-range heteronuclear techniques is that they also identify quaternary carbon resonances which one cannot observe in one-bond correlated heteronuclear experiments due to their nonprotonated behaviour.

1-D VERSIONS OF 2-D NMR EXPERIMENTS

There have been various 1D versions of 2D NMR experiments [204, 205] and these have proved to be very economical for studying oligosaccharide structures because the time required is much less than that needed for a

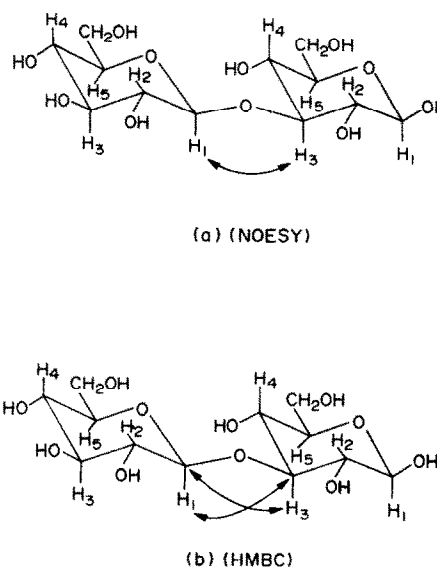


Fig. 7. Interglycosidic connectivity observed in (a) NOESY and (b) HMBC are shown schematically for the disaccharide $[\beta\text{-D-glucopyranosyl-(3}\rightarrow\text{1)-}\beta\text{-D-glucopyranose}]$.

2D experiment. These are obtained by using Gaussian-shaped pulses [206] for selective COSY experiments, combined with one- or two-step-relayed coherence transfer which provides relevant information of structural determination [207]. Semiselective excitation of one carbohydrate proton, combined with multistep-relayed coherence transfer and a terminal NOE transfer has been used for the sequential analysis of oligosaccharides [208]. A combination of the DANTE selective excitation method [209] and chemical shift selective filters [210] has also been applied to record clean 1D COSY, RELAY and NOE spectra of oligosaccharides [211].

Selected 1D TOCSY experiments are generally sufficient to obtain a complete ^1H NMR subspectrum with high digital resolution and a complete assignment of all ^1H signals of the moiety. In an analogous manner, the linkage of different sugar moieties and, hence, the assignment of the obtained subspectra to the different carbohydrate rings, can be very effectively derived by a few selected 1D NOE experiments [212].

LOCATION OF APPENDED GROUPS

The location of any appended groups if present, for example, in an acylated substituent of an acylated glycoside and an aglycone in the case of glycosides cannot be deduced from COSY or HOHAHA spectra because such groups do not form part of a J -network. However, the presence of such substituents alters the ^1H and ^{13}C NMR chemical shifts of the sugar moiety to which it is attached; thus, the site of substitution of such groups can be located.

Site of an acyl substituent. The presence of an acyl substituent, in addition to the introduction of relevant ^1H and ^{13}C resonances characteristic for such groups, causes chemical shift changes in the ^1H and ^{13}C NMR spectrum of the sugar residue. The shifts associated with such modification of the hydroxyl substituents are termed acylation-induced shifts (AIS).

In ^1H NMR spectrum, the carbinyl methine protons (protons geminal to an acyloxy substituent) appear at lowest field (4.9–6.0 ppm) with characteristic multiplicity depending upon coupling patterns. Their differentiation from the anomeric protons is usually straightforward because they usually appear as a triplet or a diffused triplet, depending upon the coupling constants with vicinal protons, whereas anomeric protons appear as a doublet with characteristic coupling constants depending upon the anomeric configuration. An exception to this behaviour is the 6-*O*-acyl derivatives in which the H-6s, in most cases, become overlapped with other ring protons of the sugar moiety.

The presence of an acyl substituent causes significant downfield shifts of the H- α -resonance. The downfield shift (α -effect) depends upon the nature of the α -carbon. Usually, the methine resonance exhibits large deshielding in the range 1.0–1.6 ppm, whereas methylene protons exhibit deshielding ranges between 0.3–0.6 ppm. The chemical shift of α -protons are dependent on the number and location of axial substituents and the proximity of the anomeric centre [213, 214]. The H-6 and H-6' protons are less deshielded (*ca.* 0.5 ppm) and the sum of the two downfield shifts are usually smaller than those for any secondary methine resonance. The deshielding of protons in β -positions is *ca.* 0.20 ± 0.5 ppm.

Acylation of an alcoholic hydroxyl group causes de-

shielding of an α -carbon by 0.6–3.5 ppm and shielding of the adjacent carbon resonance (the β -carbons) by 1.2–4.0 ppm [215–218]. An α - and β - effect of similar magnitude have also been noted for acylated glycosides [213, 214, 219]. Smaller effects on γ (–0.8 to 0.3 ppm) and δ -carbons (–0.3 to 0.3 ppm) are also observed. These acylation effects depend upon the position of *O*-acylation and the stereochemistry of the sugar residue. Systematic studies have been performed on monoacetylated methyl α - and β -gluco- and galactopyranosides which indicate that the presence of an axial substituent causes further deshielding of *ca.* 1.0 ppm, except for those in a 1,4- relationship [214]. The α -effect of acylation is *ca.* 1.0 ppm lower at the 2-position due to its proximity to the ring oxygen. These acylation-induced shifts hold reasonably well when acylation occupies the 3, 4 or 6-positions, but a diminished α -effect and, in some cases, a marginal opposite effect has been noted for 2-acylated monosaccharides. However, a β -upfield shift has always been noticed which provides valid proof for the determination of the site of acylation [220, 221]. The additive nature of these shifts has also been reported for polyacylated sugar derivatives [214, 219].

Moreover, to confirm proposed linkages, NOESY and HMBC experiments are equally important in an analogous manner as discussed for the determination of the interglycosidic linkage. The proton at the site of attachment usually exhibits a three-bond correlation to the appended group proton or vice versa. The appended groups usually exhibit a three-bond ^1H – ^{13}C coupling with the sugar resonance proving the site of acylation, therefore, ^1H – ^{13}C correlation observed in a long-range correlated spectrum confirms the substitution site. The position of acyl substituents in a flavonoid glycoside [221] and the site of acylation in a feruloylated arabinoxylan [222] was determined by NOESY experiment, whereas HMBC proved the site of acylation in an acylated steroid glycoside [223].

The position of the sulphate group in sulphated glycosides [224] and oligosaccharides can be determined because the presence of the sulphate group influences ^1H and ^{13}C NMR chemical shifts. The sulphate group causes deshielding of vicinal and geminal protons. The deshielding effect on these protons depends upon the nature of the hydroxyl group. Methylene protons show a downfield shift of 0.4–0.5 ppm, whereas methine proton exhibit a large downfield shift of 0.7–0.9 ppm. The deshielding β -effect of a primary sulphate group appears to be stronger than that of a secondary sulphate group because it leads to a downfield shift of 0.25 ± 0.05 ppm in the case of the former but 0.13 ± 0.05 ppm in the case of the latter. In ^{13}C NMR, spectra the presence of a sulphate group causes a pronounced downfield shift of 6–11 ppm of the α -resonance while a β -effect is usually shifted upfield [225]. Thus, a comparison of ^1H and ^{13}C NMR data of sulphated and unsulphated oligosaccharide can lead to the determination of the position of the sulphate group.

Site of glycosidation in glycosides. Glycosidation of a hydroxyl group, depending upon its nature (alcoholic, carboxylic or phenolic), causes a change in chemical shifts, particularly at the site of glycosylation and some other nearby resonances, that the rest of the resonances remain almost unaffected. These effects have been well studied in ^{13}C NMR spectra.

Glycosidation shifts for alcohols are similar to those of oligosaccharides, i.e. alcoholic carbons shift to lower field

by 5–12 ppm along with a concurrent upfield shift of 1.2–6.0 ppm. The intrinsic values of α - and β - effects depend upon the absence or presence of substituents at the β -position, which introduces steric hindrance. In the case of sterically unhindered secondary alcoholic glycosides, the α - effect varies between 5–8 ppm and shows dependance upon its stereochemical relationship to the pyranose ring oxygen. The β -effect is always larger (ca 4 ppm) for the *anti*-related β -resonance in comparison to the β -resonance which occupies the *syn* position because it shows an upfield shift of ca 2 ppm. In the case of sterically hindered secondary alcoholic glycosides, the α -effect is usually greater (8–12 ppm) and the β -upfield shift effect is relatively lower (1.2–3.0 ppm) or almost negligible or absent depending upon the magnitude of substitution. Usually quaternary-type β -carbons either show little or almost negligible upfield shifts and in some cases, small downfield shifts, whereas upfield shifts of β -CH₂ resonances and downfield shifts of α -resonances are reliable enough to decide about the glycosidation site [226–229].

Glycosylation of a carboxylic group causes a downfield shift of the carboxylic resonance (2–5 ppm) along with an upfield shift (0.5–2.0 ppm) of the adjacent resonance [230]. A characteristic feature of carboxylic glycosides is the appearance of the anomeric carbon of the carboxylic group-bound sugar residue at remarkably upfield position (93–97 ppm), thus identifying the sugar residue involved in glycosylation of the carboxylic group [229].

Glycosylation of a phenolic hydroxyl group causes an upfield shift of the *ipso* carbon, and a lowfield shift of resonances occupying the *ortho* and *para* positions. These effects, particularly for flavonoid *O*- and *C*-glycosides, have been recently discussed [41]; they are applicable to other naturally occurring phenolic glycosides. Interestingly these are proved also by HMBC data [201, 231]. In case of discrepancies, 1D selective INEPT, 2D NOESY and HMBC can provide further proof for the establishment of glycosidic linkages to aglycones.

CONCLUSION

The first step in the structural elucidation of an oligosaccharide is the recognition of NMR signals belonging to closed spin systems, i.e. to individual residues, and this can be best achieved with the aid of correlated spectroscopy (COSY). Starting with a given anomeric proton, one can assign the other protons which belong to the same monosaccharide unit; consequently, from their chemical shift and coupling data, the structure of the monosaccharide can be elucidated. The success of this method depends on the fact that protons of each monosaccharide constitute a separate spin system. However, carbohydrate protons which are strongly coupled, i.e. appearing in a small chemical shift range lower than 0.1 ppm lead to cross peaks hidden by cluster peaks of the diagonal. As a result, it is impossible to define the connectivity paths. In such cases, two-dimensional phase-sensitive shift correlation spectroscopy (PS-COSY), or phase-sensitive DQF-COSY, are particularly useful, because the fine structure of the peaks can be exploited to determine coupling constants of the protons involved. In addition, it is possible to discriminate between active and passive coupling by in-phase and anti-phase splitting which appears in the crosspeak as positive and negative signals. Active coupling defines the coupling between the

protons connected by a crosspeak, while passive coupling defines the coupling of one of these protons with any other. Thus, they can be used, not only for identifying sugar moieties but also for the determination of anomeric configuration. The assignment based upon COSY can be proved by TOCSY or HOHAHA or vice versa.

Intraresidual NOE effects by laboratory frame (NOESY) or rotating frame (ROESY) are useful in determining anomeric configuration, i.e., for α -anomeric configuration (NOE contracts, H-1, H-2) and for β -anomeric configuration NOE contracts, H-1, H-3 and H-5, whereas interresidue contracts establish interglycosidic linkages and the position of appended groups.

Assignment of ¹³C resonances can be achieved by considering literature values of model compounds and it can be accomplished independently via ¹H detected [¹H, ¹³C] one-bond correlation experiments. ¹H resonance assignments achieved by employing the above homonuclear techniques can readily be correlated with ¹³C frequency in HMQC experiments. Once the individual resonances have been assigned to specific sugar residues, then sites of glycosidic linkage can be established by employing glycosylation-induced shifts for ¹H and ¹³C resonances. In cases of ambiguity, these can be confirmed by means of ¹H-detected multiple bond correlation spectra optimised for long-range coupling (²J_{CH} and ³J_{CH} = ca 7 Hz) or by searching for NOEs between each of the anomeric protons, and the relevant protons of the adjacent glycosidically-linked sugar residue in a one- or two-dimensional NOE (NOESY) spectrum.

For an oligosaccharide containing identical sugar residues in only slightly different environments, the interpretation of the resonances may not be easy due to overlap problems. In such situations, sequence determination based upon interresidue NOE and long-range homo- and heteronuclear correlation may become equivocal. Combination of the above-mentioned approaches and comparison with reference data on a series of related compounds is than very effective.

Compared to conventional methods, NMR offers several advantages. It is a non-destructive technique, which can be applied to samples of the order of few milligrams in their native form, and performed in a reasonable time without degradation or chemical modification. An obvious limitation of this technique, however, concerns the determination of absolute configuration which must be achieved by conventional procedures.

It is strongly recommended that the workers in natural products chemistry should routinely use homo- and heteronuclear correlated methods not only for new compounds but also for any known compounds for which complete ¹H and ¹³C chemical shift assignments have not been previously made and to assign unambiguously ¹H and ¹³C NMR chemical shifts based on above-described and any new techniques including three-dimensional NMR [232–237]. The combination of ¹H chemical shifts and ¹H–¹H chemical shifts and ¹H–¹H coupling information along with ¹³C chemical shifts are ideally suited for complete structural analysis of naturally occurring oligosaccharides and glycosides.

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